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Master's Degree in Quantum Computing

The anatomy of quantum
kernels: how circuit de-
sign shapes classification
and concentration

Master's Thesis

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Resumen

Los métodos basados en kernels cuánticos dependen en gran medida del diseño del circuito utilizado como mapa de características cuántico, aunque todavía no existe un criterio claro para determinar qué arquitecturas resultan más adecuadas para tareas de clasificación. Este Trabajo analiza de forma sistemática la influencia de distintas decisiones de diseño sobre el comportamiento de estos modelos.

Para ello se estudian cuatro aspectos complementarios: la relación entre las propiedades estructurales de los circuitos y el rendimiento en clasificación, el efecto de las puertas de codificación y del entrelazamiento mediante estudios de ablación, el impacto del número de qubits y la relación entre la arquitectura del circuito, la geometría del embedding cuántico y la estructura del kernel. Los experimentos se realizan sobre cuatro problemas sintéticos de clasificación utilizando máquinas de vectores soporte con kernels cuánticos basados en fidelidad.

Los resultados muestran que propiedades globales como la expresividad o la capacidad de entrelazamiento, consideradas de forma aislada, no explican de manera consistente el rendimiento observado. En cambio, el comportamiento de los kernels cuánticos depende de la interacción entre múltiples decisiones de diseño, incluyendo la estrategia de codificación, la arquitectura del circuito y el tamaño del sistema. En conjunto, el trabajo proporciona una evaluación controlada de estos factores y ofrece una visión más detallada de cómo influyen en el comportamiento de los mapas de características cuánticos.

Palabras clave: computación cuántica, aprendizaje automático cuántico, kernel cuántico, mapas de características cuánticos, máquinas de vectores soporte.

Abstract

Quantum kernel methods rely heavily on the design of the quantum circuit used as a quantum feature map, yet there is still no clear criterion for selecting circuit architectures for classification tasks. This Work presents a systematic study of how different circuit design choices influence the behaviour of quantum kernel models.

Four complementary aspects are investigated: the relationship between structural circuit properties and classification performance, the effect of encoding gates and entanglement through controlled ablation studies, the impact of the number of qubits, and the connection between circuit architecture, quantum embedding geometry, and kernel structure. The evaluation is carried out on four synthetic classification datasets using support vector machines with fidelity-based quantum kernels.

The results indicate that global circuit descriptors such as expressibility and entangling capability, when considered in isolation, do not consistently explain the observed classification performance. Instead, the behaviour of quantum kernels is determined by the interaction of multiple design choices, including the encoding strategy, circuit architecture, and system size. Overall, this Work provides a controlled evaluation of these factors and offers further insight into how they shape the behaviour of quantum feature maps.

Keywords: quantum computing, quantum machine learning, quantum kernels, quantum feature maps, support vector machines.

1. Introduction

This Chapter presents the motivation and context for this Work. It traces the development of machine learning and quantum computing as converging disciplines, identifies the open questions that this Work addresses, and describes the structure of the document.

1.1 Motivation

Across domains as diverse as image recognition (Krizhevsky et al., 2012), natural language processing (Vaswani et al., 2017), and medical diagnosis (Esteva et al., 2017), the rules governing the underlying patterns are either unknown or too complex to be explicitly formalized (LeCun et al., 2015). It was Arthur Samuel (1959) who coined the term machine learning (ML) and first gave it operational meaning: programming a computer to behave “in a way which, if done by human beings or animals, would be described as involving the process of learning”. Rather than encoding rules by hand, the system would learn them from experience. The key insight is that, given enough representative examples, a system can generalize — that is, perform correctly on inputs it has never seen before (Goodfellow et al., 2016). This intuition was formalized into a rigorous statistical framework (Mitchell, 1997; Vapnik, 1998), establishing the theoretical conditions under which such generalization is possible and giving rise to the discipline that now underpins much of modern artificial intelligence (LeCun et al., 2015).

In the 1980s, Rumelhart et al. (1986) demonstrated the effectiveness of backpropagation for training neural networks with hidden layers, enabling them to learn non-linear representations. Despite this progress, these networks remained difficult to train (Glorot & Bengio, 2010) and offered limited theoretical guarantees on generalization (Vapnik, 1998). Against this backdrop, Boser et al. (1992) introduced Support Vector Machines (SVMs), a learning method grounded in Vapnik’s statistical framework. SVMs find the decision boundary that maximizes the separation between classes — a principle with direct theoretical justification in terms of generalization (Vapnik, 1998) — and, through the so-called kernel trick, operate implicitly in high-dimensional feature spaces without ever computing the transformation explicitly (Boser et al., 1992; Cortes & Vapnik, 1995). This made SVMs both theoretically principled and practically powerful. Their limitation, however, lies elsewhere: the choice of kernel is critical, and designing kernels that faithfully capture

the structure of complex, high-dimensional data remains a non-trivial problem (Chapelle et al., 2002).

Although SVMs remained highly successful in many applications, research increasingly shifted towards representation learning with neural networks. AlexNet (Krizhevsky et al., 2012) marked a turning point in the field: deep neural networks, trained on large-scale datasets using GPUs, reduced the error rate in the ImageNet competition by nearly 10 percentage points over the previous state of the art, demonstrating that scale and depth could unlock capabilities that had long seemed out of reach (LeCun et al., 2015). This momentum culminated in the introduction of the Transformer architecture (Vaswani et al., 2017), whose attention mechanism allowed models to capture complex relationships in data more effectively than any previous approach. Scaled to billions of parameters and trained on large datasets, Transformer-based models have since become the dominant paradigm in machine learning, achieving strong performance across language, vision, and scientific domains (Brown et al., 2020).

Despite this progress, classical ML faces fundamental limitations in certain problem domains (Marcus, 2018). The computational cost of training large models grows at a pace that is difficult to sustain (Strubell et al., 2019), and in scientific domains such as chemistry (Troyer & Wiese, 2005) or materials science (Kohn, 1999), the complexity of the underlying systems grows exponentially with their size — making classical representations increasingly inadequate (Feynman, 1982). More broadly, even well-established methods such as kernel-based SVMs depend critically on the choice of kernel, and designing kernels that faithfully capture the structure of complex data remains an open problem (Schölkopf & Smola, 2002). These limitations suggest that progress on certain classes of problems may require computational paradigms that go beyond classical information processing.

At the beginning of the 20th century, classical physics failed to explain several phenomena at the subatomic scale (Griffiths & Schroeter, 2018). Quantum mechanics emerged as the theoretical framework that could, and over the following decades it became one of the most successful and carefully tested theories in the history of science (Planck, 1901; Einstein, 1905; Dirac, 1930). Once this framework was firmly established, Feynman (1982) observed that simulating quantum systems on a classical computer carries an exponential cost, and proposed that only a computer operating under quantum mechanical principles could do so efficiently.

D. Deutsch (1985) was the first to formalize a theoretical model of quantum compu-

tation, demonstrating that a quantum computer could solve a specific problem — determining whether a function is constant or balanced — in fewer steps than any classical algorithm. This was an early instance of what is now called quantum advantage: the ability of a quantum computer to solve certain problems with substantially fewer computational resources than any known classical approach (Nielsen & Chuang, 2010). Shor (1994) then showed that a quantum computer could factor large integers into their prime factors in polynomial time, an exponential improvement over the best known classical algorithms and a result with direct implications for modern cryptography. Shortly after, Grover (1996) demonstrated a quadratic quantum advantage for unstructured search problems, showing that quantum speedup was not an isolated phenomenon but a broader computational principle. Together, these results established quantum computing as a serious research field.

Although quantum advantage has been theoretically demonstrated for specific problems, current quantum hardware operates in what Preskill (2018) termed the Noisy Intermediate-Scale Quantum (NISQ) era: devices with a limited number of qubits that are highly sensitive to noise and decoherence, restricting the practical implementation of algorithms such as Shor’s or Grover’s. Within these constraints, a natural research direction emerged: rather than waiting for fault-tolerant quantum computers, hybrid classical-quantum algorithms were developed in which a quantum device handles specific computational steps while a classical computer manages the rest (Cerezo et al., 2021). This hybrid paradigm is the foundation on which quantum machine learning (QML) is built — a field that explores how quantum computing can enhance or extend classical machine learning methods (Biamonte et al., 2017).

The theoretical appeal of QML lies in the ability of quantum systems to process information in ways that have no efficient classical equivalent (Biamonte et al., 2017). Within this field, several approaches have been proposed, ranging from quantum versions of neural networks to hybrid optimization algorithms (Cerezo et al., 2021). However, many of these approaches face practical training difficulties that limit their applicability (McClean et al., 2018). Quantum kernel methods offer a more theoretically grounded alternative: rather than training a quantum circuit directly, they use quantum systems to define similarity measures between data points, inheriting the well-established theoretical framework of classical kernel methods (Havlíček et al., 2019; Schuld, 2021).

Quantum kernel methods are, however, not without limitations. Their practical perfor-

mance depends critically on how the quantum system is designed, yet there is no general consensus on which architectural choices — such as the type of gates, the entangling structure, or the circuit depth — lead to better learning performance.

1.2 Problem statement

Quantum kernel methods offer a theoretically principled approach to machine learning: by encoding classical data into quantum states and computing similarity through quantum circuits, they inherit the well-established framework of kernel-based SVMs while operating in exponentially large feature spaces (Havlíček et al., 2019; Schuld & Killoran, 2019). However, the practical behavior of these methods is poorly understood. The performance of a quantum kernel depends on the specific circuit used to encode the data — its gate composition, entanglement structure, and depth — yet there is no established understanding of how these choices translate into classification performance or kernel quality.

This Work approaches that question empirically. Starting from a systematic evaluation of quantum circuits varying in entanglement pattern, depth, and number of qubits, it examines whether architectural choices produce measurable differences in classification performance across multiple binary classification datasets. As circuits scale, the analysis progressively focuses on the structural properties of the resulting kernel matrices, investigating the conditions under which exponential concentration emerges and characterizing its relationship with circuit design. Throughout, quantum kernel classifiers are compared against classical baselines to contextualize the results within the broader landscape of kernel methods.

1.3 Document structure

The remainder of this document is organized as follows.

- **Chapter 2** introduces the theoretical background required to understand this Work: supervised learning, support vector machines, and kernel methods from a classical perspective, followed by the fundamentals of quantum computing and quantum machine learning, and concluding with the evaluation metrics used throughout the study.
- **Chapter 3** states the general and specific objectives of this Work.

- **Chapter 4** describes the experimental methodology, including the circuit architectures evaluated, the datasets used, and the experimental protocol.
- **Chapter 5** presents and analyzes the experimental results.
- **Chapter 6** discusses the main findings and their implications.
- **Chapter 7** outlines directions for future research.

2. Context and state of the art

This Chapter provides the theoretical background required to understand this Work. It is organized into four sections. The first covers the classical machine learning concepts that underpin quantum kernel methods, including supervised learning, support vector machines, and kernel methods. The second introduces the fundamental concepts of quantum computing. The third presents quantum machine learning, with particular focus on data encoding, quantum kernels, and their known advantages and limitations. The fourth introduces the evaluation metrics used throughout the experimental study.

2.1 Classical background

This Section introduces the classical machine learning concepts that underpin this Work. Starting from the fundamentals of supervised learning and generalization, it develops the theory of Support Vector Machines and kernel methods.

2.1.1 Supervised learning

Classical computers work with bits, units of information that take values 0 or 1 (Shannon, 1948). All data — numbers, text, images — is stored and manipulated as sequences of bits through algorithms: finite sequences of well-defined instructions that transform an input into an output (Cormen et al., 2022). The design of such algorithms requires that the rules governing the task be known and explicitly encoded. However, for many real-world problems — such as recognizing objects in images (Krizhevsky et al., 2012) or diagnosing diseases from medical records (Rajkomar et al., 2018) — the underlying rules are either unknown or too complex to formalize manually (LeCun et al., 2015). Machine learning addresses this limitation by inferring a mapping $f : \mathcal{X} \rightarrow \mathcal{Y}$ directly from data (Mitchell, 1997; Shalev-Shwartz & Ben-David, 2014), where $\mathcal{X}$ represents the space of possible inputs (for example, images or patient records) and $\mathcal{Y}$ the space of possible outputs (for example, object categories or diagnoses), shifting the burden from rule engineering to learning from examples.

Given a dataset $\mathcal{D}$, it is split into two subsets: the training set $\mathcal{D}_{train}$, used to learn the structure of the problem, and the test set $\mathcal{D}_{test}$, used to evaluate the performance of the model on unseen data (Shalev-Shwartz & Ben-David, 2014). In supervised learning, each

subset consists of input samples X and their corresponding labels Y , where each sample x_i is paired with a label y_i . The model learns a function $f : X_{train} \rightarrow Y_{train}$ and is then evaluated on its ability to predict labels for new samples in X_{test} .

The central goal of supervised learning is generalization: the ability to perform well on data not seen during training (Shalev-Shwartz & Ben-David, 2014). Formally, this means minimizing the error the model would make on any data it could encounter, not just the error measured on the training examples themselves (Vapnik, 1998). A model that fits the training data too closely may fail to generalize — a phenomenon known as overfitting — while a model that is too simple may fail to capture the underlying structure of the data, leading to underfitting (Bishop, 2006). Balancing these two extremes is a central challenge in machine learning, and one that directly motivates the design of Support Vector Machines.

2.1.2 Support Vector Machines

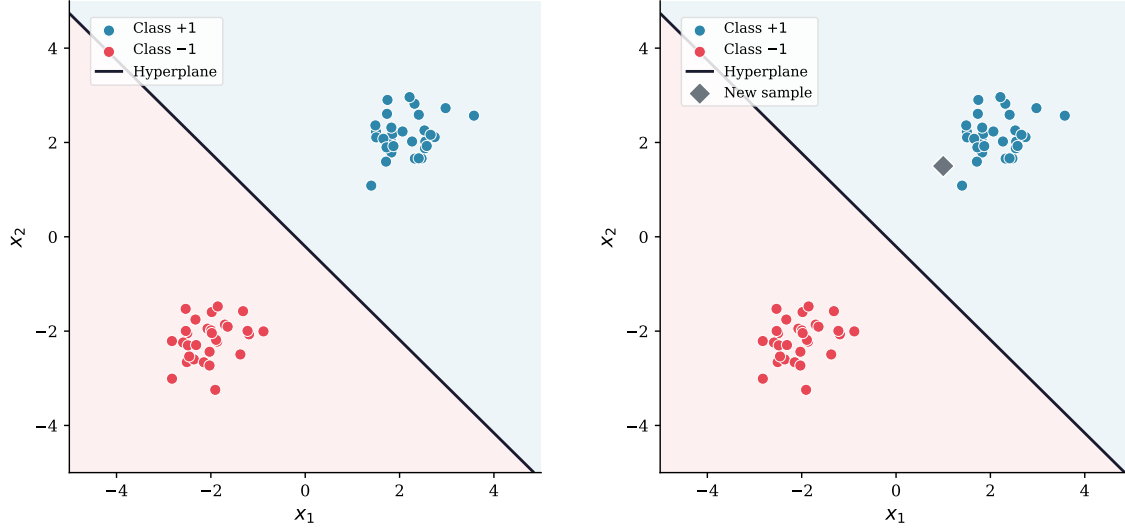
Given a binary dataset $\mathcal{D} = \{(x_i, y_i)\}$ where $y_i \in \{-1, +1\}$ are the class labels, the goal of binary classification is to find a function f that assigns the correct label to new, unseen samples. A Support Vector Machine (SVM) (Cortes & Vapnik, 1995) is a supervised learning method that finds a decision boundary separating the two classes. In a d -dimensional space, this boundary is a $(d - 1)$ -dimensional hyperplane: in two dimensions it is a line, in three dimensions a plane, and so on.

As an example, consider the two-dimensional dataset shown in Figure 2.1a, where each point x_i has two features and its color indicates its label y_i . The hyperplane divides the space into two regions — one for each class — and the sign of the decision function determines the predicted label for any point. When a new, unseen sample x^* is received, as shown in Figure 2.1b, the SVM predicts its label based on which side of the hyperplane it lies on. Formally, given a sample x , the decision function is:

$$f(x) = w^\top x + b$$

where $w \in \mathbb{R}^d$ is the weight vector normal to the hyperplane, $b \in \mathbb{R}$ is the bias term, and the predicted label is:

$$\hat{y} = \text{sign}(f(x)) = \begin{cases} +1 & \text{if } f(x) > 0 \\ -1 & \text{if } f(x) < 0 \end{cases}$$



(a) Training samples and separating hyperplane.

(b) Classification of a new unseen sample.

Figure 2.1: Illustration of binary classification using a linear decision boundary. Source: own work.

In linearly separable datasets, many different hyperplanes may correctly separate the two classes. The key idea of SVM is not only to find a separating hyperplane, but to select the one that maximizes the distance to the nearest training points of each class (Boser et al., 1992). This distance is known as the margin and the nearest points are called the support vectors.

To find the optimal hyperplane, the SVM solves the following optimization problem:

$$\min_{w,b} \frac{1}{2} \|w\|^2$$

subject to the constraints:

$$y_i(w^T x_i + b) \geq 1, \quad \forall i$$

where (x_i, y_i) are the training samples and labels with $y_i \in \{-1, +1\}$. Since the margin is inversely proportional to the norm of the normal vector w , maximizing the margin is equivalent to minimizing $\|w\|^2$. A more detailed representation of this optimization process is shown in Figure 2.2, where the margin boundaries and the support vectors are highlighted.

Since the margin is the distance between the hyperplane and the nearest training points of each class, we can derive it formally. The distance from a point x_i to the hyperplane

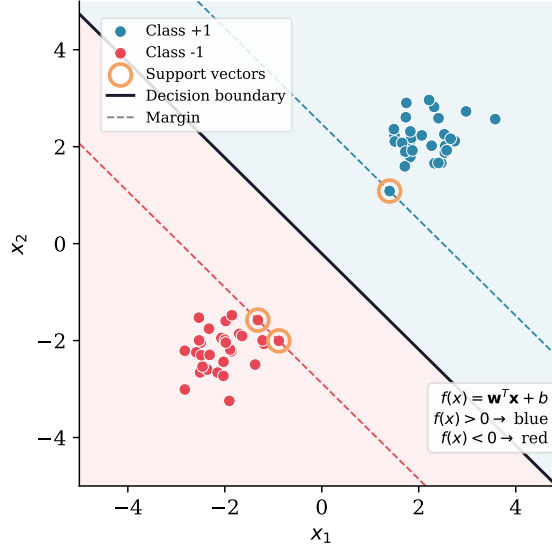


Figure 2.2: SVM with linearly separable data. Source: own work.

defined by $w^\top x + b = 0$ is given by:

$$d(x_i) = \frac{|w^\top x_i + b|}{\|w\|}$$

The constraints $y_i(w^\top x_i + b) \geq 1$ imply that the support vectors — the closest points to the hyperplane — satisfy $y_i(w^\top x_i + b) = 1$, so their distance to the hyperplane is exactly $\frac{1}{\|w\|}$. Since there is one such point on each side, the total margin between the two classes is:

$$\text{margin} = \frac{2}{\|w\|}$$

Maximizing this margin is therefore equivalent to minimizing $\|w\|^2$, which is the objective of the optimization problem stated above. This formulation is known as the hard-margin SVM, as it assumes the data to be perfectly linearly separable and allows no misclassifications. In practice, however, data is rarely linearly separable, which motivates two complementary extensions of the basic SVM formulation.

There is an alternative formulation for cases where the data is mostly linearly separable but contains some degree of overlap or noise near the decision boundary. In such situations, no hyperplane can separate both classes without misclassifying a few points, as illustrated in Figure 2.3a. The soft-margin SVM (Cortes & Vapnik, 1995) addresses this by allowing some points to violate the margin constraints, trading a small number of misclassifications for a more robust decision boundary.

This is formalized by introducing slack variables $\xi_i \geq 0$, one for each training point,

which measure the degree of constraint violation. The optimization problem becomes:

$$\min_{w,b,\xi} \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \xi_i$$

subject to:

$$y_i(w^\top x_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0, \quad \forall i$$

The parameter $C > 0$ controls the trade-off between maximizing the margin and minimizing constraint violations. A large C penalizes misclassifications heavily, resulting in a narrower margin that fits the training data more closely. A small C allows more violations, producing a wider margin at the cost of more training errors. As shown in Figure 2.3b, the soft-margin SVM captures the underlying linear structure of the data even in the presence of noisy or overlapping points.

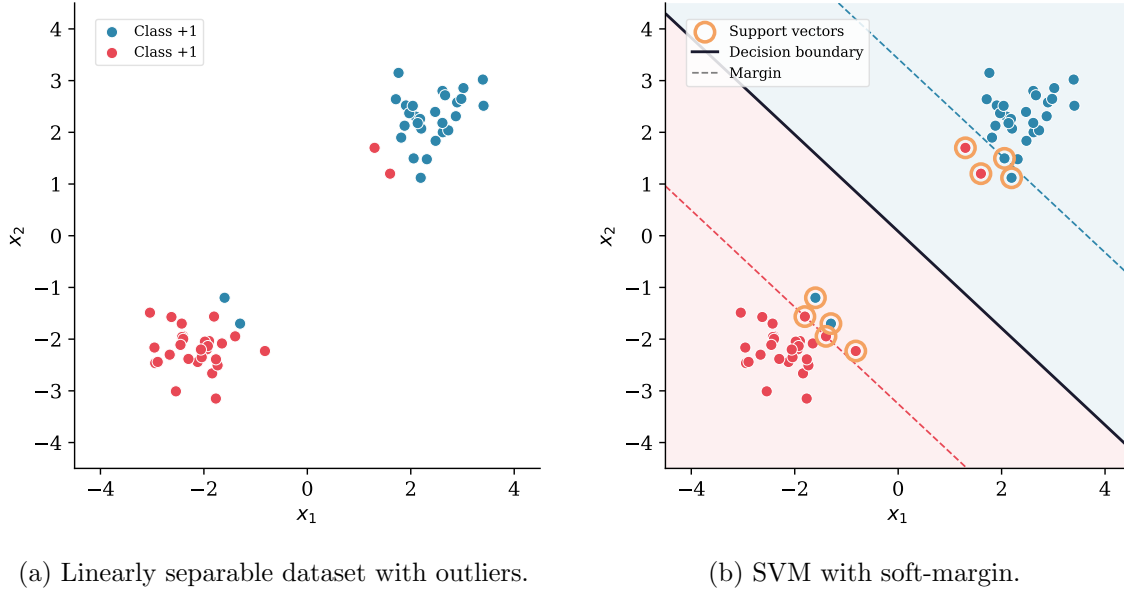


Figure 2.3: Illustration of binary classification using a linear decision boundary with soft-margin. Source: own work.

Soft-margin SVM handles noise and mild overlap, but it does not address a more fundamental limitation: data whose structure is inherently non-linear. In Figure 2.4, the data follows a circular structure that is clearly non-linearly separable: no straight line can correctly divide the two classes, regardless of how many misclassifications are permitted. This is not a matter of noise or overlapping points — the classes are perfectly distinguishable, but their boundary is inherently non-linear. In such cases, relaxing the margin constraints is insufficient; a fundamentally different approach is needed.

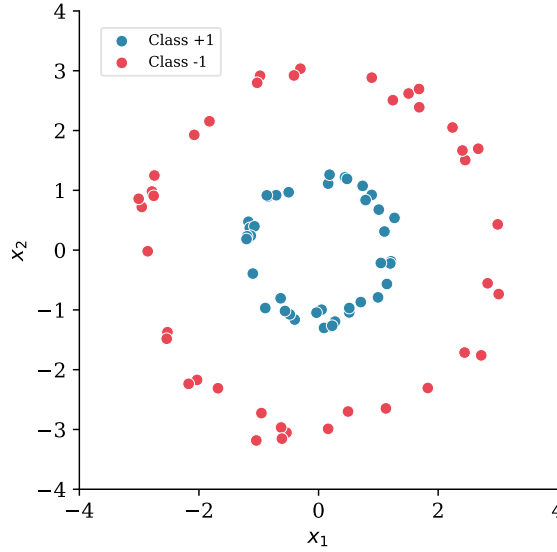


Figure 2.4: Example of a non-linearly separable dataset. Source: own work.

The key insight is that, although the data is not linearly separable in the original space, it may become linearly separable if mapped to a higher-dimensional space. Intuitively, adding dimensions can allow a hyperplane to correctly separate classes that overlap in the original space. This idea is the foundation of kernel methods.

2.1.3 Kernel methods

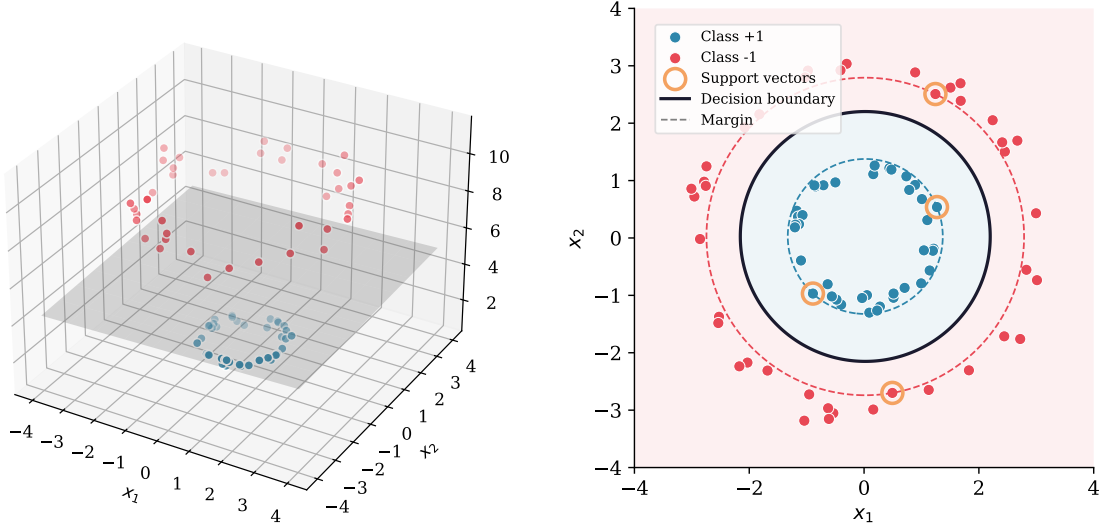
Kernel methods address the problem of non-linearly separable data by mapping it into a higher-dimensional feature space (Schölkopf & Smola, 2002), where a linear separation becomes possible. This is achieved through a transformation function ϕ :

$$\phi : \mathbb{R}^d \rightarrow \mathbb{R}^{d'}, \quad d' > d$$

In this transformed space, a linear decision boundary such as a hyperplane can be used to separate the data. As a concrete example, consider the following mapping from $\mathbb{R}^2$ to $\mathbb{R}^3$:

$$\phi(x_1, x_2) = (x_1, x_2, x_1^2 + x_2^2)$$

The data shown in Figure 2.4, distributed in circular patterns, is not linearly separable in its original two-dimensional space. After applying the transformation above, the data is mapped into three dimensions, as shown in Figure 2.5a. In this new feature space, the two classes can be easily separated by a linear decision boundary such as a plane, as shown in Figure 2.5b.



(a) Non-separable data in two dimensions mapped to three dimensions, where it becomes linearly separable.

(b) Linear decision boundary in the transformed feature space, corresponding to a non-linear boundary in the original space.

Figure 2.5: Effect of a feature map on a non-linearly separable dataset. Source: own work.

However, computing $\phi(x)$ explicitly can be computationally expensive or even infeasible when the target feature space is very high-dimensional or infinite. Furthermore, it can be shown that the SVM optimization problem admits a dual formulation (Kuhn & Tucker, 1951; Boser et al., 1992; Cortes & Vapnik, 1995) in which the solution depends on the training data only through pairwise inner products $\langle x_i, x_j \rangle$. As a consequence, the decision function can be written as:

$$f(x) = \sum_{i=1}^n \alpha_i y_i \langle x_i, x \rangle + b$$

where $\alpha_i \geq 0$ are the dual coefficients, non-zero only for the support vectors. This is a key observation: the SVM never needs the data points themselves, only their pairwise similarities measured through inner products (Schölkopf & Smola, 2002). If instead of computing inner products in the original space one could compute them in the transformed space $\langle \phi(x_i), \phi(x_j) \rangle$ — without ever constructing ϕ explicitly — the SVM would operate in the higher-dimensional feature space at no additional cost. This is precisely what a kernel function achieves, and it is called the kernel trick: rather than computing $\phi(x)$ explicitly, a kernel function directly evaluates the inner product in the transformed space (Aizerman, 1964; Boser et al., 1992):

$$k(x_i, x_j) = \langle \phi(x_i), \phi(x_j) \rangle$$

This makes it possible to work efficiently in high-dimensional or even infinite-dimensional feature spaces using only kernel evaluations, without ever constructing ϕ explicitly. Returning to the donut example, the mapping $\phi(x_1, x_2) = (x_1, x_2, x_1^2 + x_2^2)$ corresponds to the kernel:

$$k(x, x') = x_1 x'_1 + x_2 x'_2 + (x_1^2 + x_2^2)(x_1'^2 + x_2'^2)$$

which can be evaluated directly without computing ϕ at all.

Not all functions k are valid kernels. For a function to qualify as a kernel, it must be symmetric, that is $k(x_i, x_j) = k(x_j, x_i)$, and positive semi-definite: the kernel matrix K with entries $K_{ij} = k(x_i, x_j)$ must satisfy $v^\top K v \geq 0$ for all vectors $v \in \mathbb{R}^n$. By Mercer's theorem, these conditions are equivalent to the existence of a feature map ϕ such that $k(x_i, x_j) = \langle \phi(x_i), \phi(x_j) \rangle$, guaranteeing that the kernel corresponds to a valid inner product in some feature space (Mercer, 1909; Schölkopf & Smola, 2002). This ensures that the SVM optimization problem remains well-defined when inner products are replaced by kernel evaluations.

The kernel matrix $K \in \mathbb{R}^{n \times n}$, defined as:

$$K_{ij} = k(x_i, x_j)$$

gathers all pairwise similarity information between training points. It is the structure the SVM operates on: the decision function in the dual formulation becomes:

$$f(x) = \sum_{i=1}^n \alpha_i y_i k(x_i, x) + b$$

where $\alpha_i \geq 0$ are the dual coefficients, non-zero only for the support vectors. The geometry of K directly reflects the capability of the kernel to separate classes: a well-designed kernel produces a matrix with clear block structure, with high values between points of the same class and low values between points of different classes (Cristianini et al., 2001). A kernel that fails to capture the structure of the data produces a nearly uniform matrix, making classification unreliable.

Several kernel functions have been proposed in the literature, each implicitly defining a different feature space. The linear kernel is the simplest case, corresponding to the standard inner product in the original input space:

$$k(x_i, x_j) = x_i^\top x_j$$

It does not perform any transformation, so it is equivalent to a standard linear SVM. It is most effective when the data is already linearly separable or approximately so (Schölkopf & Smola, 2002).

The polynomial kernel of degree d maps the data to a space of polynomial features up to degree d :

$$k(x_i, x_j) = (x_i^\top x_j + c)^d$$

where $c \geq 0$ is a constant that controls the influence of lower-degree terms. It captures interactions between features at multiple scales, making it suitable for problems where such structure is present (Boser et al., 1992; Schölkopf & Smola, 2002).

The radial basis function (RBF) kernel, also known as the Gaussian kernel, is one of the most widely used in practice:

$$k(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2)$$

where $\gamma > 0$ controls the width of the Gaussian. Unlike the previous two, the RBF kernel implicitly maps data into an infinite-dimensional feature space, giving it great flexibility to capture complex non-linear boundaries. However, it is sensitive to the choice of γ : large values lead to narrow kernels that may overfit, while small values produce smoother decision boundaries at the risk of underfitting (Broomhead & Lowe, 1988; Cortes & Vapnik, 1995; Schölkopf & Smola, 2002).

The performance of the SVM depends critically on the kernel choice, but it is not always trivial to find the right one.

2.2 Quantum computing

This Section introduces the fundamental concepts of quantum computing required to understand quantum kernel methods. Unlike classical bits, quantum systems operate under the principles of quantum mechanics, allowing forms of information processing with no classical analogue. The concepts introduced here — qubits, quantum gates, entanglement, and measurement — form the language in which quantum circuits and quantum kernels are defined in the following section.

2.2.1 Qubits and quantum states

In classical computers, a bit can take the value 0 or 1. A system of two bits can represent two independent binary values, resulting in $2^2 = 4$ possible configurations, but only one

can be active at a time. More generally, n classical bits can represent 2^n configurations, but only one at any given moment.

Quantum bits, or qubits, can also take the values 0 or 1, represented by the quantum states $|0\rangle$ and $|1\rangle$ in Dirac notation (Dirac, 1930; Nielsen & Chuang, 2010), but they can also exist in a superposition of both states simultaneously. Quantum superposition refers to the ability of a quantum system to exist simultaneously in a linear combination of multiple basis states until a measurement is performed (Nielsen & Chuang, 2010).

A qubit is described by the quantum state:

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$$

where $\alpha, \beta \in \mathbb{C}$ are complex amplitudes whose squared magnitudes represent the probabilities of measuring the qubit in each basis state, satisfying the normalization condition $|\alpha|^2 + |\beta|^2 = 1$ (Nielsen & Chuang, 2010).

For a system of two qubits, the state is described by a superposition of all four basis states (Nielsen & Chuang, 2010):

$$|\psi\rangle = \alpha_{00} |00\rangle + \alpha_{01} |01\rangle + \alpha_{10} |10\rangle + \alpha_{11} |11\rangle, \quad \sum_{i,j \in \{0,1\}} |\alpha_{ij}|^2 = 1$$

More generally, a system of n qubits exists in a 2^n -dimensional Hilbert space $\mathcal{H}$ — a complex vector space with an inner product that allows distances and angles between quantum states to be defined (von Neumann, 1932; Nielsen & Chuang, 2010). While a classical system can only occupy one of 2^n configurations at any given moment, a quantum system can simultaneously represent a superposition of all 2^n basis states. This exponential growth in the dimensionality of the state space is what makes quantum computing particularly appealing for machine learning tasks where capturing complex structure in high-dimensional spaces is crucial (Biamonte et al., 2017).

2.2.2 Quantum gates and circuits

Quantum gates are the basic operations that manipulate qubit states. Analogously to classical logic gates, they transform an input quantum state into an output quantum state. Mathematically, quantum gates are represented by unitary matrices U satisfying $UU^\dagger = I$, which guarantees that the normalization condition on quantum states is preserved (Nielsen & Chuang, 2010). A sequence of quantum gates applied to one or more qubits forms a quantum circuit — the standard model of quantum computation (Nielsen & Chuang, 2010).

Single-qubit gates

Single-qubit gates act on one qubit at a time, rotating its state on the Bloch sphere (D. E. Deutsch, 1989; Nielsen & Chuang, 2010) — a geometric representation of the qubit state space in which the north and south poles correspond to $|0\rangle$ and $|1\rangle$ respectively, and any point on the surface represents a valid qubit state (Nielsen & Chuang, 2010).

The Pauli gates are the three fundamental single-qubit gates, each corresponding to a rotation of π radians around one of the axes of the Bloch sphere (Nielsen & Chuang, 2010). The Pauli- X gate is the quantum analogue of the classical NOT gate: it flips the qubit state, mapping $|0\rangle \mapsto |1\rangle$ and $|1\rangle \mapsto |0\rangle$. The Pauli- Y gate applies a combined bit and phase flip. The Pauli- Z gate leaves $|0\rangle$ unchanged and applies a phase flip to $|1\rangle$. Their matrix representations are:

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \quad Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Beyond their role as gates, the Pauli matrices also serve as the standard observables for quantum measurement, as discussed in Section 2.2.4.

The Hadamard gate maps the computational basis states to equal superpositions:

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle), \quad H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

and is represented by the matrix (Nielsen & Chuang, 2010):

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

It is one of the most widely used gates in quantum computing, as it provides a simple way to create superposition from a classical basis state.

The rotation gates R_X , R_Y and R_Z generalize the Pauli gates by rotating the qubit state around the X , Y and Z axes of the Bloch sphere by a continuous angle θ (Nielsen & Chuang, 2010):

$$R_X(\theta) = \exp(-i\frac{\theta}{2}X) = \begin{bmatrix} \cos(\theta/2) & -i\sin(\theta/2) \\ -i\sin(\theta/2) & \cos(\theta/2) \end{bmatrix}$$

$$R_Y(\theta) = \exp(-i\frac{\theta}{2}Y) = \begin{bmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{bmatrix}$$

$$R_Z(\theta) = \exp(-i\frac{\theta}{2}Z) = \begin{bmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{bmatrix}$$

When $\theta = \pi$, each rotation gate is equivalent to its corresponding Pauli gate up to a global phase. These gates are particularly relevant for data encoding in quantum machine learning, as classical values can be embedded into quantum states by using them as rotation angles (Schuld, 2021).

As a concrete example, consider a qubit initialized in $|0\rangle$, corresponding to the north pole of the Bloch sphere, as shown in Figure 2.6. Applying $R_Y(\theta)$ rotates the state on the XZ plane. Figures 2.7a–2.7d illustrate how the qubit state evolves under increasing values of θ . For $\theta = \pi/2$:

$$R_Y\left(\frac{\pi}{2}\right) |0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = |+\rangle$$

which places the qubit on the equator of the Bloch sphere, in an equal superposition of both basis states — the same state produced by the Hadamard gate applied to $|0\rangle$.

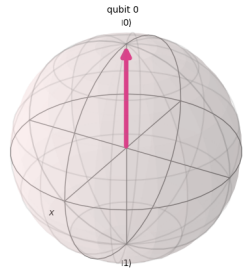


Figure 2.6: Qubit in the $|0\rangle$ state on the Bloch sphere. Source: own work.

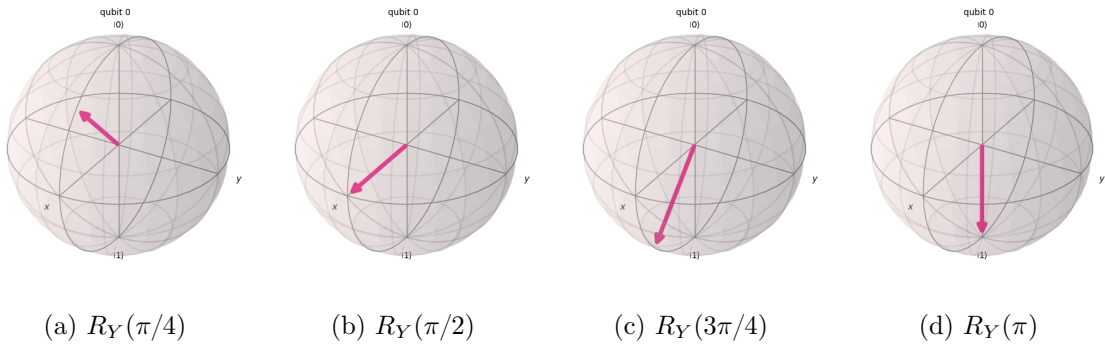


Figure 2.7: Evolution of the qubit state on the Bloch sphere under different $R_Y(\theta)$ rotations. Source: own work.

Two-qubit gates

Two-qubit gates act on a pair of qubits simultaneously, introducing correlations between them — a quantum correlation called entanglement, explained in Section 2.2.3. The most common structure is the controlled gate: one qubit acts as the control and the other as the target. The operation is applied to the target only when the control qubit is in the state $|1\rangle$, leaving it unchanged otherwise (Nielsen & Chuang, 2010).

The CNOT gate (Controlled-NOT) is the most fundamental two-qubit gate (Barenco et al., 1995). It flips the target qubit if and only if the control qubit is in $|1\rangle$:

$$\text{CNOT}|00\rangle = |00\rangle, \quad \text{CNOT}|01\rangle = |01\rangle, \quad \text{CNOT}|10\rangle = |11\rangle, \quad \text{CNOT}|11\rangle = |10\rangle$$

Its matrix representation in the computational basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$ is:

$$\text{CNOT} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

Beyond CNOT, a family of controlled rotation gates extends this idea to parametrized rotations: a $CR_k(\theta)$ gate applies a rotation $R_k(\theta)$ around axis $k \in \{X, Y, Z\}$ to the target qubit conditioned on the control qubit being in $|1\rangle$. These gates introduce tunable entangling operations and are widely used in parameterized quantum circuits (Sim et al., 2019).

Quantum circuits

A quantum circuit is an ordered sequence of quantum gates applied to a register of n qubits. The system is typically initialized in the state $|0\rangle^{\otimes n}$, and the circuit defines the complete unitary transformation from this initial state to the output state on which a measurement is performed (Nielsen & Chuang, 2010). The expressiveness of a quantum circuit — its ability to represent a rich variety of quantum states — depends on the choice of gates, their arrangement, and the parameters they use (Sim et al., 2019).

2.2.3 Entanglement

Quantum entanglement is a property of certain multi-qubit states that cannot be written as a product of individual qubit states (Schrödinger, 1935; Bell, 1964; Horodecki et al.,

2009; Nielsen & Chuang, 2010):

$$|\psi\rangle_{AB} \neq |\psi\rangle_A \otimes |\psi\rangle_B$$

Such states are called entangled, and they have no classical analogue. A canonical example is the Bell state $|\Phi^+\rangle$, one of the four maximally entangled Bell states (Bell, 1964):

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

This state can be prepared by applying a Hadamard gate to the first qubit followed by a CNOT gate:

$$\text{CNOT}(H \otimes I) |00\rangle = \text{CNOT} \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) |0\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

In $|\Phi^+\rangle$, neither qubit has a definite state on its own — the system exists as a whole. As a consequence, measuring one qubit instantly determines the outcome of measuring the other: if the first qubit is found in $|0\rangle$, the second is also $|0\rangle$, and similarly for $|1\rangle$, regardless of any physical separation between them (Nielsen & Chuang, 2010).

In the context of quantum circuits, entanglement arises whenever a two-qubit gate such as CNOT acts on a qubit in superposition. Without entanglement, each qubit evolves independently and the joint state of the system remains separable. With entanglement, the circuit can represent correlations between qubits that have no product-state description, giving access to a strictly richer region of the Hilbert space (Nielsen & Chuang, 2010).

2.2.4 Measurement

Measurement is the process by which a quantum state is observed, producing a classical outcome. When a qubit in state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ is measured, the superposition collapses irreversibly to one of the basis states: the outcome is 0 with probability $|\alpha|^2$ and 1 with probability $|\beta|^2$ (von Neumann, 1932; Nielsen & Chuang, 2010). After measurement, all information about the original superposition is lost — the qubit is now in a definite classical state.

This probabilistic nature has an important practical consequence: a single measurement of a quantum circuit provides only one sample from the output distribution. To estimate expectation values reliably, the circuit must be executed many times and the results averaged (Nielsen & Chuang, 2010).

Measurement bases and observables

The description above assumes measurement in the computational basis $\{|0\rangle, |1\rangle\}$, which corresponds to measuring whether the qubit is aligned with the Z axis of the Bloch sphere. More generally, a qubit can be measured with respect to any observable — a Hermitian operator O whose eigenvalues are the possible measurement outcomes (von Neumann, 1932; Nielsen & Chuang, 2010). The expected value of an observable O on a state $|\psi\rangle$ is (Nielsen & Chuang, 2010):

$$\langle O \rangle = \langle \psi | O | \psi \rangle$$

The Pauli matrices X , Y and Z , introduced in the previous section as quantum gates, are the standard choice of observables in quantum computing. Each corresponds to measuring the qubit along a different axis of the Bloch sphere:

- Z measurement collapses the qubit to $|0\rangle$ or $|1\rangle$, with outcomes $+1$ and -1 respectively.
- X measurement collapses the qubit to $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ or $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$, with outcomes $+1$ and -1 respectively.
- Y measurement collapses the qubit to $|+i\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle)$ or $|-i\rangle = \frac{1}{\sqrt{2}}(|0\rangle - i|1\rangle)$, with outcomes $+1$ and -1 respectively (where $i = \sqrt{-1}$).

2.3 Quantum machine learning

Quantum machine learning (QML) is a research field that explores how quantum computing can enhance or extend classical machine learning methods (Biamonte et al., 2017). A central challenge in QML is bridging the gap between classical data and quantum computation: since quantum circuits operate on quantum states, classical input data must first be encoded into a quantum representation before any quantum processing can take place.

2.3.1 Data encoding

There are different strategies to encode classical data into quantum states. In general, encoding is achieved by applying a parametrized quantum circuit that prepares a quantum state whose structure depends on the input data. Formally, a classical vector $x \in \mathbb{R}^d$ is mapped to a quantum state $|\psi(x)\rangle$ via a unitary operation $U(x)$ acting on the initial state

$|0\rangle^{\otimes n}$:

$$|\psi(x)\rangle = U(x) |0\rangle^{\otimes n}$$

The function $U(x)$ is called the quantum feature map, and different choices of U lead to different embeddings of the data in Hilbert space (Schuld, 2021). Three of the most widely used encoding strategies are described below (Schuld & Petruccione, 2018):

- **Basis encoding** represents discrete data values directly as computational basis states (Trugenberger, 2001). For example, the integer 3 (binary 011) is encoded as $|011\rangle$. This method is straightforward but does not exploit quantum superposition and is generally limited to discrete inputs.
- **Amplitude encoding** embeds data values into the amplitudes of a quantum state (Long & Sun, 2001; Plesch & Brukner, 2011). For a normalized vector $x \in \mathbb{R}^n$:

$$|\psi(x)\rangle = \sum_{i=0}^{n-1} x_i |i\rangle, \quad \sum_i |x_i|^2 = 1$$

This encoding is compact, requiring only $\log_2(n)$ qubits to represent n features, but it requires complex state preparation circuits that are generally difficult to implement on near-term hardware.

- **Angle encoding** maps each feature to the rotation angle of a single-qubit gate, such as $R_Y(x_i)$. Each feature alters the qubit's position on the Bloch sphere, producing a state that geometrically reflects the input value (Mitarai et al., 2018). This method is hardware-efficient and straightforward to implement, at the cost of requiring one qubit per feature.

A crucial observation is that the quantum feature map is not determined solely by the encoding strategy: the full circuit — its depth, gate composition, and entangling structure — shapes the final state $|\psi(x)\rangle$ in Hilbert space. Two circuits using the same encoding method but different architectures will in general produce different embeddings, leading to different decision boundaries and kernel behaviors. This makes the design of the quantum feature map a central concern in QML, and one of the main questions investigated in this Work.

Once classical data has been encoded as quantum states, a natural question arises: how can the similarity between two encoded states be quantified? This is precisely the role of quantum kernels, introduced in the following subsection.

2.3.2 Quantum kernels

Parameterized quantum circuits can be used to define kernel functions by acting as quantum feature maps that embed classical data into a high-dimensional Hilbert space (Schuld & Killoran, 2019). Given a feature map implemented by a unitary $U(x)$, the quantum state associated with an input x is:

$$|\psi(x)\rangle = U(x) |0\rangle^{\otimes n}$$

The quantum kernel is then defined as the fidelity between two such states:

$$k(x, x') = |\langle \psi(x) | \psi(x') \rangle|^2$$

This measures the overlap between the quantum states corresponding to two data points — a natural notion of similarity in Hilbert space (Havlíček et al., 2019). Using the unitarity of $U(x)$, this can be rewritten as:

$$k(x, x') = |\langle 0 | U^\dagger(x') U(x) | 0 \rangle|^2$$

which provides a practical circuit-based procedure for computing the kernel: prepare $U(x) |0\rangle$, apply $U^\dagger(x')$, and measure the probability of returning to $|0\rangle$ (Buhrman et al., 2001; Schuld & Killoran, 2019; Havlíček et al., 2019). The resulting kernel matrix $K_{ij} = k(x_i, x_j)$ can then be used directly in a support vector machine, replacing the classical kernel in the dual formulation.

This formulation shows that the quantum kernel is entirely determined by the choice of feature map — that is, by the structure of the quantum circuit $U(x)$. Different circuit architectures induce different notions of similarity in the input space, and therefore different kernel matrices and classification boundaries. In particular, the arrangement of single-qubit rotations and entangling operations directly shapes the geometry of the embedding in Hilbert space.

An alternative to the fidelity kernel is the projected quantum kernel (PQK), introduced by Huang et al. (2021). Rather than measuring the overlap between full quantum states, PQKs project each state onto local observables and construct a classical kernel from those projections. This approach is more robust to concentration effects and was shown to retain predictive power in regimes where fidelity kernels collapse. While this Work focuses on fidelity-based kernels, PQKs represent an important reference point in the landscape of quantum kernel methods.

2.3.3 Advantages and limitations of quantum kernels

Quantum kernels provide a principled extension of classical kernel methods to quantum feature spaces. When a fixed quantum kernel is used within a support vector machine, the resulting training problem retains the convex optimization properties of classical SVMs (Schölkopf & Smola, 2002; Havlíček et al., 2019). Their main appeal lies in the fact that quantum feature maps can embed classical data into Hilbert spaces whose dimension grows exponentially with the number of qubits. In addition, for certain feature maps, estimating the corresponding kernel values may be classically intractable, which is generally considered a prerequisite for potential quantum advantage (Tang, 2019; Havlíček et al., 2019; Liu et al., 2021). Under carefully constructed assumptions, a rigorous quantum advantage over classical learners has indeed been proven (Liu et al., 2021), although the learning problem considered there was specifically designed to exhibit this separation and its practical relevance remains unclear.

In practice, however, translating these theoretical properties into improved learning performance remains challenging. Huang et al. (2021) showed that the structure of the training data can strongly affect whether theoretical quantum advantages persist in realistic settings, since classical learners may exploit sample distributions to approximate target functions efficiently. Likewise, empirical benchmarking studies have often found that quantum kernel classifiers do not systematically outperform well-established classical baselines on standard classification tasks, suggesting that favorable performance is highly problem-dependent (Bowles et al., 2024).

At a more fundamental level, Thanasilp et al. (2024) showed that quantum kernel values can exponentially concentrate around a constant as the system size increases, causing the kernel matrix to become nearly uniform and reducing the model’s sensitivity to the input data. In a related study on variational quantum classifiers, Hubregtsen et al. (2021) found that expressibility correlated strongly with classification accuracy, while entangling capability showed only a weak correlation, indicating that these structural descriptors do not uniformly predict performance. Together, these results suggest that the relationship between circuit design and learning performance — whether measured through kernel quality or classification accuracy — remains poorly understood across quantum machine learning paradigms. Determining which architectural choices lead to informative kernels, and which instead induce concentration or poor generalization, therefore remains an open

research question.

2.4 Evaluation metrics

Evaluating the quality of a kernel requires looking beyond classification performance. The geometry of the kernel matrix itself carries information about whether the kernel is capturing meaningful structure in the data or collapsing toward an uninformative representation. This Section introduces the metrics used throughout this Work to assess both aspects.

2.4.1 Classification metrics

The most direct measure of kernel quality is the performance of the classifier it induces. Since quantum kernels are used within a support vector machine, classification performance is measured through accuracy — the proportion of correctly classified samples:

$$\text{Accuracy} = \frac{\text{number of correct predictions}}{\text{total number of samples}}$$

Two variants are reported throughout this Work: training accuracy, computed on the samples used to fit the classifier, and test accuracy, computed on a held-out set not seen during training. The gap between the two is informative: a large gap signals overfitting, while low accuracy on both sets indicates that the kernel fails to capture the structure of the data.

2.4.2 Kernel metrics

The following metrics characterize the structure and discriminative power of the kernel matrix.

Off-diagonal variance

The off-diagonal variance is the variance of all non-diagonal entries of the kernel matrix:

$$\sigma_{\text{off}}^2 = \text{Var}(\{K_{ij} : i \neq j\})$$

When a kernel is fully concentrated, all off-diagonal entries converge to the same value and this variance approaches zero (Thanasilp et al., 2024). A high off-diagonal variance indicates that the kernel assigns meaningfully different similarity values to different pairs of points, which is a necessary condition for useful classification.

Effective dimension

The effective dimension quantifies how many eigenvalues contribute meaningfully to the kernel representation, using the participation ratio of the eigenspectrum (Gao et al., 2017):

$$d_{\text{eff}} = \frac{(\sum_{i=1}^n \lambda_i)^2}{\sum_{i=1}^n \lambda_i^2}$$

This quantity lies in the range $[1, n]$. When the spectrum is dominated by a single eigenvalue, $d_{\text{eff}} \approx 1$, indicating that the kernel effectively operates in a one-dimensional subspace. When all eigenvalues are equal, $d_{\text{eff}} = n$, indicating that representational capacity is uniformly distributed across all available dimensions.

Support vector fraction

The fraction of support vectors is the proportion of training points that are selected as support vectors by the SVM, i.e., the number of support vectors divided by the total number of training samples. In a well-generalizing model, the decision boundary is determined by a sparse subset of critical points near the margin, yielding a low fraction. Conversely, when the kernel fails to capture meaningful structure in the data, the SVM requires nearly every training point to define the hyperplane, driving the fraction toward unity. A high support vector fraction therefore serves as a direct indicator of overfitting and poor generalization, complementing the generalization gap as a diagnostic for kernel collapse (Cortes & Vapnik, 1995).

2.4.3 Feature map metrics

While kernel metrics describe the similarity structure induced by the feature map, feature map metrics focus on the properties of the resulting quantum states in Hilbert space.

Shannon entropy

For a quantum state $|\psi(x)\rangle$, the probability distribution over the computational basis is:

$$p_z(x) = |\langle z | \psi(x) \rangle|^2$$

The Shannon entropy of this distribution is (Shannon, 1948):

$$H(x) = - \sum_z p_z(x) \log p_z(x)$$

A state concentrated in a single basis element has $H = 0$. A state uniformly distributed over all 2^n basis states achieves the maximum entropy $H_{\max} = \log(2^n) = n \log 2$. In the results, entropy is reported normalized by this maximum:

$$\tilde{H}(x) = \frac{H(x)}{H_{\max}}$$

so that $\tilde{H} \in [0, 1]$ regardless of the number of qubits. As the system size grows and the embedding becomes more expressive, each encoded state tends to spread more uniformly over the exponentially large basis — a behavior directly linked to kernel concentration, since states that are each individually close to uniform will also be close to each other in fidelity.

Participation ratio

The participation ratio (PR) of an encoded quantum state $|\psi(x)\rangle$ quantifies the effective number of computational basis states $\{|z\rangle\}$ that contribute to its superposition, a concept originating in the study of localization in disordered systems (Anderson, 1958; Evers & Mirlin, 2008). Mathematically, it is defined as the reciprocal of the purity of the state's population distribution:

$$PR(x) = \frac{1}{\sum_z p_z(x)^2}$$

where $p_z(x) = |\langle z|\psi(x)\rangle|^2$ represents the probability of measuring the state $|\psi(x)\rangle$ in the basis element $|z\rangle$, subject to the normalization condition $\sum_z p_z(x) = 1$.

Rather than a discrete count of basis states with non-zero probability, the PR acts as a continuous measure of distribution homogeneity. Because the probabilities are squared in the denominator, the metric is heavily biased toward dominant components. High-probability weights dominate the sum, while low-probability components are suppressed quadratically. Consequently, the PR reflects the effective dimensionality spanned by the state's weights, positioning its values strictly within the range $[1, 2^n]$. A state fully localized on a single basis element yields $PR = 1$, whereas a completely deslocalized state—characterized by a uniform probability distribution ($p_z = 1/2^n$)—reaches the upper bound $PR = 2^n$.

This metric is conceptually analogous to the effective dimension defined in Section 2.4, but operates at a different hierarchy of the quantum pipeline. While the effective dimension evaluates the spectral flatness of the collective kernel matrix to gauge its global

representational capacity, the participation ratio monitors the geometric spreading of individual data points. Together, they provide complementary views of the same concentration phenomenon: an escalation in the average participation ratio towards 2^n signals that individual states are collapsing into uniform noise, which fundamentally drives the flattening of the kernel eigenvalues and the subsequent loss of generalization.

Class centroid distance

The class centroid distance measures the separation between the two classes in the amplitude space of the embedding, extending the classical centroid-based notion of class separation used in pattern recognition (Bishop, 2006). For each data point x , the amplitude vector is defined as $\phi(x) = (|\langle z | \psi(x) \rangle|)_z \in \mathbb{R}^{2^n}$. The class centroids are:

$$c_k = \frac{1}{|S_k|} \sum_{x \in S_k} \phi(x), \quad k \in \{0, 1\}$$

and the centroid distance is $d_c = \|c_0 - c_1\|_2$. A larger centroid distance corresponds to greater separation between the class centroids in the amplitude space, whereas smaller values indicate that the centroids are closer together. This metric provides a geometric measure of class separation in the embedding space.

2.4.4 Spearman correlation

Finally, Spearman's rank correlation coefficient ρ is also reported when comparing empirical performance rankings against theoretical rankings from previous work (Spearman, 1904). Unlike the previous measures, which quantify absolute properties of the kernel or feature map, Spearman's ρ assesses the degree of monotonic agreement between two ordered rankings. This makes it particularly suitable for evaluating whether the relative ordering of circuits induced by theoretical criteria (e.g., expressibility or entangling capability) is consistent with the ordering induced by classification performance. Statistical significance is assessed through the associated p -value, which quantifies the probability of observing such a correlation under the null hypothesis of no association.

$$\rho = \text{corr}(\text{rank}(X), \text{rank}(Y))$$

3. Objectives

This Work investigates the role of quantum circuit architecture in determining the behavior of quantum kernel methods for binary classification. The study is structured around two interconnected questions: whether architectural choices systematically affect classification performance, and under which conditions the resulting kernels exhibit exponential concentration. The following objectives guide the empirical investigation.

3.1 General objectives

- To analyze how quantum circuit design choices — including gate type, entanglement structure, circuit depth, and number of qubits — influence the classification performance of quantum kernel methods.
- To investigate how quantum circuit architecture shapes the geometry of the induced quantum feature space and the resulting kernel matrices.
- To characterize the phenomenon of exponential kernel concentration and identify the architectural factors that contribute to its emergence as circuits scale.

3.2 Specific objectives

- To implement and evaluate a systematic grid of quantum circuits varying in entanglement pattern (none, linear, full), number of qubits, and number of encoding layers across multiple binary classification datasets.
- To quantify classification performance using training and test accuracy, and to analyze the generalization gap between them, enabling a structured comparison across circuit configurations.
- To characterize kernel concentration and the geometry of the quantum feature space through complementary kernel-level and feature-map-level metrics, analyzing how these quantities evolve with circuit depth and system size.
- To determine how different entanglement topologies influence kernel quality and classification performance, and to assess the relative contribution of circuit depth versus qubit count to concentration effects.

4. Methodology

This Chapter describes the experimental methodology followed in this Work. It covers the encoding strategy used to embed classical data into quantum circuits, the circuit architectures evaluated, the datasets used, and the experimental protocol followed to obtain the results presented in Chapter 5.

4.1 Encoding

Throughout this Work, quantum kernels are constructed from parameterized quantum feature maps that encode classical data into Hilbert space. Most experiments employ a custom family of feature maps specifically designed for this study, while Experiment 4 additionally considers the standard ZZ feature map (Havlíček et al., 2019).

As introduced in Chapter 2, classical data can be embedded into a quantum Hilbert space in several ways. This Work employs angle-based feature maps, in which classical features are encoded through data-dependent rotation angles. In the custom circuits developed for Experiments 4.5-4.7, each feature of the input vector is assigned to the rotation angle of a single-qubit gate, so that a feature value $x_i \in \mathbb{R}$ is encoded as $R_k(x_i)$ for some rotation axis k . Experiment 4.8 employs the standard ZZ feature map implementation, which also belongs to the family of angle-based feature maps, although it combines single-qubit rotations with predefined data-dependent phase interactions between qubits.

Within this strategy, there is still a design choice: how many features to assign per qubit. One option is to apply multiple rotation gates to a single qubit, encoding several features through different axes. But since the objective of this Work is to isolate the effect of circuit architecture rather than study encoding strategies, each qubit is assigned exactly one feature. This direct encoding strategy is simple, easily interpretable, and has been widely adopted in the quantum kernel literature (Havlíček et al., 2019).

A quantum circuit used as a feature map can contain two types of gates: encoding gates, whose rotation angles depend on the input data point x , and structural gates, whose parameters are fixed and identical for every input. This distinction is essential. The quantum kernel is computed as:

$$k(x, x') = |\langle 0 | U^\dagger(x') U(x) | 0 \rangle|^2$$

which corresponds to applying $U(x)$ to $|0\rangle^{\otimes n}$, then the inverse circuit $U^\dagger(x')$, and measuring the probability of returning to $|0\rangle$. The kernel therefore evaluates a single combined circuit $U^\dagger(x')U(x)$, whose purpose is to measure how distinguishable the two encodings are. Only encoding gates contribute to this comparison: since structural gates have the same parameters for every input, a structural gate V in $U(x)$ and its inverse $V^\dagger$ in $U^\dagger(x')$ always appear as adjacent pairs and cancel exactly, contributing nothing to the kernel value.

As a concrete example, consider a single-qubit circuit $U(x) = R_Z(\theta) R_X(x)$, where $R_X(x)$ is an encoding gate and $R_Z(\theta)$ is a structural gate with a fixed parameter θ . The kernel circuit becomes:

$$\langle 0 | R_X(-x') R_Z(-\theta) R_Z(\theta) R_X(x) | 0 \rangle$$

The two structural rotations cancel — $R_Z(-\theta) R_Z(\theta) = I$ — and the kernel reduces to:

$$k(x, x') = |\langle 0 | R_X(-x') R_X(x) | 0 \rangle|^2$$

The structural gate has no effect whatsoever on the kernel. The same cancellation occurs for any structural gate that is not separated from its counterpart in $U^\dagger(x')$ by an encoding gate. For a structural layer to have any influence on the kernel, it must be placed between two encoding blocks — so that the encodings prevent the cancellation. A circuit with L structural layers therefore requires $L + 1$ encoding blocks.

This design principle directly determines the general structure of all circuits used in this Work: every circuit begins and ends with an encoding block, with structural blocks placed between them. The specific composition of those structural blocks varies across experiments and is described in the following Sections.

Finally, since each qubit encodes exactly one feature, the minimum number of qubits is determined by the data dimensionality. To keep this constraint simple and consistent across all experiments, all datasets used in this Work have exactly two dimensions. For circuits with more than two qubits, the encoding repeats: all even-indexed qubits encode x_1 and all odd-indexed qubits encode x_2 , allowing circuits of different sizes to be evaluated under the same encoding strategy.

4.2 Datasets

To benchmark the performance and expressivity of the quantum feature maps under different geometric decision boundaries, four distinct synthetic datasets are utilized. To ensure strict consistency and comparability across all experimental runs, several global constraints are enforced. First, as established in Section 4, all datasets are defined in a two-dimensional continuous feature space, $X \in \mathbb{R}^2$, and represent a binary classification task where each data point is assigned a label $y \in \{0, 1\}$. Second, every dataset consists of exactly $N = 400$ samples.

The underlying spatial structure of each dataset is fixed using a seed of 42 to guarantee invariance across all experiments. Regarding the train-test splits, the strategy depends on the experimental design: for experiments with random structural parameters (Experiments 1–3), the split is fixed to isolate the variance introduced by the circuit initialization; for the fully deterministic circuits in Experiment 4, the split is varied across multiple seeds to assess generalization robustness.

The spatial distribution and labeling of each dataset are depicted in Figure 4.1. The specific construction and geometric properties of each dataset are described below:

- **Moons:** Generated using the `make_moons` function from the scikit-learn library (Pedregosa et al., 2011). This dataset comprises two interleaving, crescent-shaped half-circles that are inherently non-linearly separable. A Gaussian noise component with a standard deviation of $\sigma = 0.1$ is added to the coordinates to blur the decision boundary. Geometrically, the two clusters intertwine, presenting a classic challenge for linear classifiers and demanding a feature map capable of disentangling continuous curvilinear structures.
- **Circles:** Generated using the `make_circles` function from scikit-learn (Pedregosa et al., 2011). It consists of two concentric circles in the two-dimensional plane, where one larger circle completely encloses a smaller, inner circle. A noise factor of $\sigma = 0.15$ is introduced, and the ratio between the inner and outer radius is strictly set to 0.4. This dataset is completely linearly inseparable in its original Euclidean representation and tests the capacity of the quantum feature map to perform a radial projection into a higher-dimensional space where a separating hyperplane can be found.
- **Gaussian XOR:** This dataset implements a multi-cluster quadrant-based structure

representing the exclusive-OR (XOR) logic gate problem. It is constructed by generating four independent isotropic Gaussian clusters. Two clusters centered around coordinates $(1, 1)$ and $(-1, -1)$ are assigned to Class 0, while the remaining two clusters centered at $(1, -1)$ and $(-1, 1)$ are assigned to Class 1. Each quadrant contains an equal allocation of 100 samples, governed by a standard deviation of $\sigma = 0.3$. This structure tests the feature map’s ability to identify diagonal relationships and multi-modal distributions.

- **Checkerboard:** This dataset constructs a complex 4×4 grid arrangement of alternating tiles, representing a highly non-linear and discontinuous decision boundary. Samples are uniformly drawn from a continuous bounding box spanning $[-2, 2]^2$. Because the domain covers four unit intervals along each dimension, the application of the floor function partitions the space into 16 distinct sectors. The binary label assignment is mathematically governed by the parity rule:

$$y = (\lfloor x_1 \rfloor + \lfloor x_2 \rfloor) \bmod 2$$

This dataset introduces sharp, axis-aligned step functions and complex boundary intersections, evaluating whether the quantum kernel can map intricate, localized grid patterns effectively.

Together, these datasets span a range of geometric structures — from smooth global boundaries to highly localized decision regions — allowing the robustness of the different circuit architectures to be evaluated under qualitatively different classification tasks.

4.3 Implementation framework

All experiments in this Work are implemented in Python. Quantum circuits are constructed and simulated using PennyLane (Bergholm et al., 2022), a framework for differentiable quantum programming that provides a flexible interface for defining parameterized quantum circuits and computing quantum kernel matrices. Classical machine learning components — including the support vector machine classifier, dataset generation, and train-test splitting — are handled through scikit-learn (Pedregosa et al., 2011). Numerical operations are performed with NumPy (Harris et al., 2020), and results are aggregated and stored using pandas (The pandas development team, 2020).

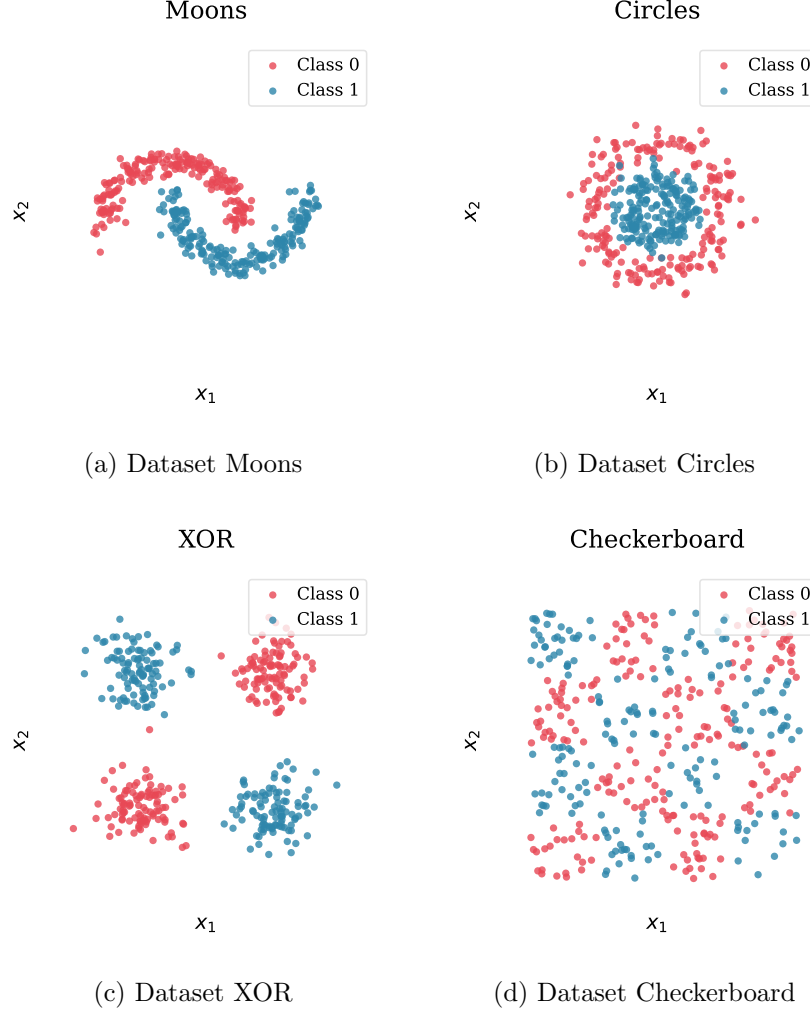


Figure 4.1: All four datasets used in the experimentation. Source: own work.

For each experimental configuration, the quantum kernel matrix is computed by evaluating the fidelity kernel $k(x_i, x_j) = |\langle 0|U^\dagger(x_j)U(x_i)|0\rangle|^2$ for all pairs of training points, and a separate cross-kernel matrix is computed between test and training points. Both matrices are passed directly to a support vector machine with a precomputed kernel, trained using scikit-learn's `SVC` with regularization parameter $C = 1.0$.

As described in Section 4.1, the custom feature maps contain structural gates whose parameters are fixed after initialization and are not learned during training. Since different random initializations may induce different feature maps, each circuit configuration is evaluated over 20 independent runs, using a different random seed to sample the structural parameters in every run. To isolate the variability introduced by the circuit itself, the dataset and its train-test split remain fixed throughout these repetitions. Consequently,

any variation observed across runs can be attributed exclusively to the random initialization of the structural gates.

Deterministic feature maps, such as the ZZ feature map used in Experiment 4, do not contain randomly initialized structural parameters. In this case, repeating the experiment with different initializations would produce identical results. Instead, the train-test split is varied across multiple random seeds to assess the robustness of the feature map with respect to the sampled training data.

4.4 Classical separability baseline

Before evaluating quantum kernel models, a preliminary baseline is established using standard classical kernels with support vector machines. The goal of this step is not to compare methods in terms of performance, but to obtain a rough reference of how linearly or non-linearly separable each dataset is. This helps to put the later quantum results into context.

Three support vector machine kernels are considered for this purpose: linear, polynomial, and radial basis function (RBF). For each of them, the hyperparameters are tuned using cross-validation over a predefined search space. In particular, the regularization parameter is optimized in all cases, while the polynomial kernel also explores different degrees and gamma values, and the RBF kernel explores different values of the kernel width parameter.

All experiments follow the same data splits described in Section 4.2, using stratified sampling to keep class proportions balanced. Cross-validation is applied only on the training set to select the best configuration for each kernel and dataset combination, and the final performance is then evaluated on the test set.

This setup provides a simple way of characterizing how each dataset behaves under increasing model flexibility, from linear decision boundaries to more complex nonlinear mappings. The resulting behavior is used as a qualitative reference when interpreting the performance of the quantum kernels later on.

4.5 Experiment 1 — Circuit architecture as a predictor of kernel performance

This experiment investigates whether two structural properties of quantum circuits — expressibility and entangling capability, as defined by Sim et al. (2019) — are reliable predictors of classification performance when those circuits are used as quantum feature maps. The following sections describe the circuit architectures evaluated, the metrics used to characterize them, and the experimental protocol followed to assess the correspondence between structural rankings and observed accuracy.

4.5.1 Motivation

A natural starting point for studying quantum kernel methods is to ask whether existing characterizations of circuit quality translate into classification performance. Sim et al. (2019) proposed two descriptors — expressibility and entangling capability — to compare parameterized quantum circuits independently of any specific algorithm, and provided rankings for a benchmark set of nineteen circuit templates evaluated on four qubits. These descriptors were originally developed for variational algorithms, but they capture structural properties — the ability to explore Hilbert space and to generate entangled states — that are equally relevant when a circuit is used as a feature map. If more expressive or more entangling circuits consistently produce better kernels, these descriptors would provide a principled criterion for circuit selection. This experiment tests that hypothesis directly by evaluating all nineteen circuits on the four classification datasets described in Section 4.2 and comparing the resulting accuracy rankings against the structural rankings reported by Sim et al. (2019).

4.5.2 Circuit architectures

The nineteen circuit templates evaluated in this experiment are taken directly from Sim et al. (2019). Each template is defined on four qubits and combines gates from the set $\{H, R_X, R_Y, R_Z, \text{CNOT}, \text{CZ}, \text{CRX}, \text{CRZ}\}$ in different arrangements. The circuits vary considerably in complexity: the simplest consist of a single layer of single-qubit rotations with no entangling gates (Circuit 1), while the most complex employ all-to-all entanglement strategies with multiple alternating rotation and entangling blocks (Circuits 5 and 6).

Entanglement topologies include nearest-neighbor, ring, and all-to-all configurations, and the entangling gates range from fixed two-qubit operations (CNOT, CZ) to parameterized controlled rotations (CRX, CRZ).

A key design decision in this experiment concerns how the encoding is integrated with each circuit template. As discussed in Section 4.1, only gates whose rotation angles depend on the input data contribute to the kernel value; structural gates with fixed parameters cancel in the kernel computation unless they are interleaved between encoding blocks. Rather than selecting a different encoding gate for each of the nineteen templates — which would conflate the effect of the encoding with that of the structural block — a uniform encoding strategy is adopted: an R_Y layer is prepended before each structural layer and appended after the last one, so that every circuit shares exactly the same encoding structure. Under this scheme, the only difference between any two circuits is the structural block itself, making the comparison fair and controlled. The rotation angles of the structural gates are sampled randomly but with fixed seeds shared across all circuits, as described in Section 4.3. The full set of circuit diagrams is shown in Figures 4.2 and 4.3.

All circuits are evaluated with a single structural layer ($L=1$), as this is the configuration for which Sim et al. (2019) provide a complete and explicit ranking of the nineteen circuit templates in terms of both expressibility and entangling capability, enabling a direct comparison with their reported values. While the original study also investigates how these descriptors evolve with circuit depth, it does not provide a full ordinal ranking for $L > 1$ that would allow the same level of systematic comparison. For each circuit, the mean test accuracy over the random seeds is computed separately for each dataset, yielding one accuracy value per circuit per dataset. Circuits are then ranked by descending accuracy within each dataset to produce the ordinal ranking used in the correlation analysis.

4.5.3 Expressibility and entangling capability

To contextualize the accuracy results, this experiment relies on the expressibility and entangling capability rankings reported by Sim et al. (2019) for the nineteen circuit templates at $L = 1$. Neither metric is recomputed here; instead, the ordinal rankings are inferred directly from Figures 3 and 4 of the original paper and used as reference orderings for the correlation analysis.

Expressibility measures how well a parameterized circuit can approximate the uniform distribution of pure states over the Hilbert space, known as the Haar distribution

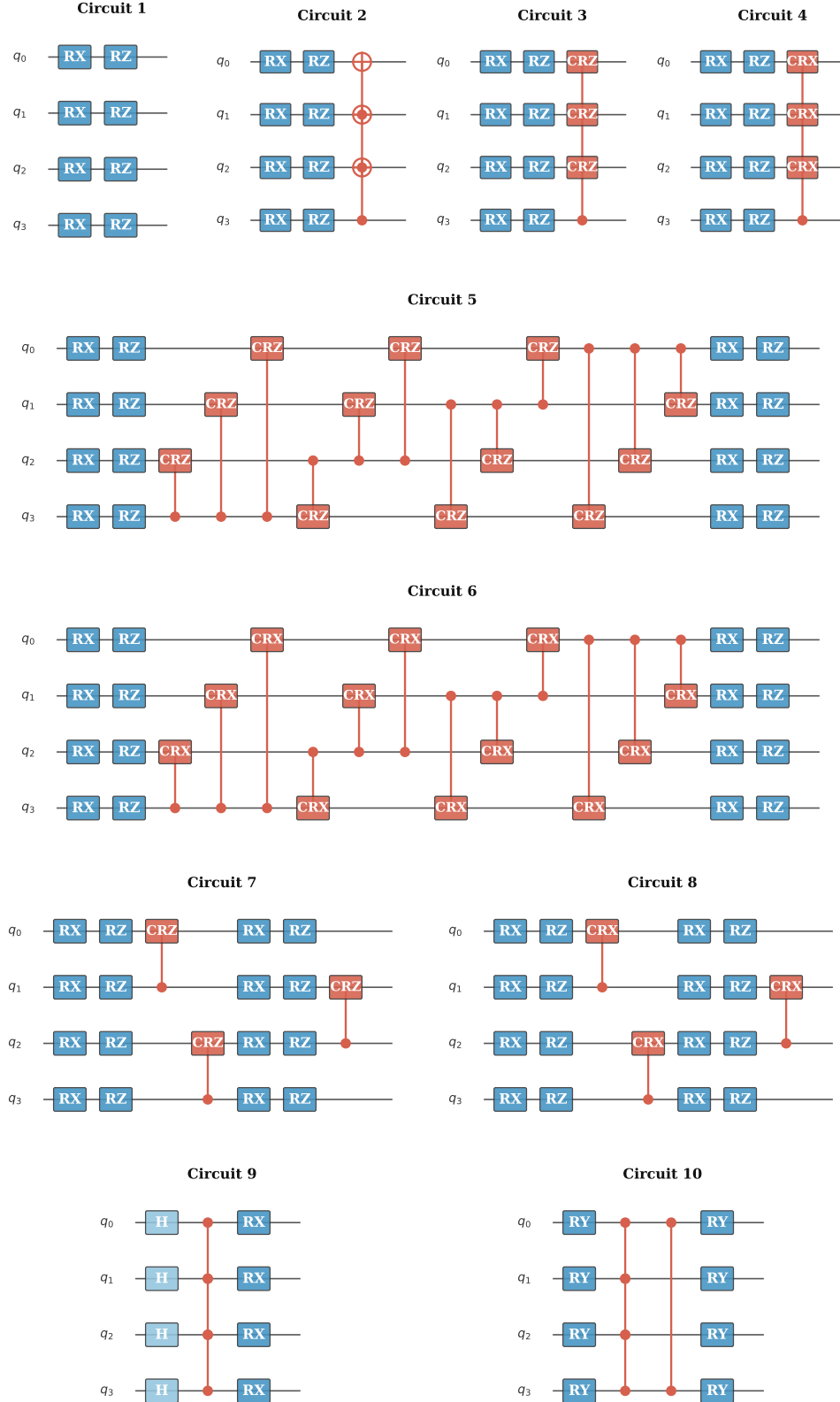


Figure 4.2: Circuit templates 1–10 from Sim et al. (2019), adapted as quantum feature maps. Blue boxes indicate single-qubit rotation gates; red elements indicate entangling gates. All circuits are shown for a single structural layer ($L = 1$) and four qubits. Encoding R_Y blocks are not shown. Source: adapted from Sim et al. (2019)

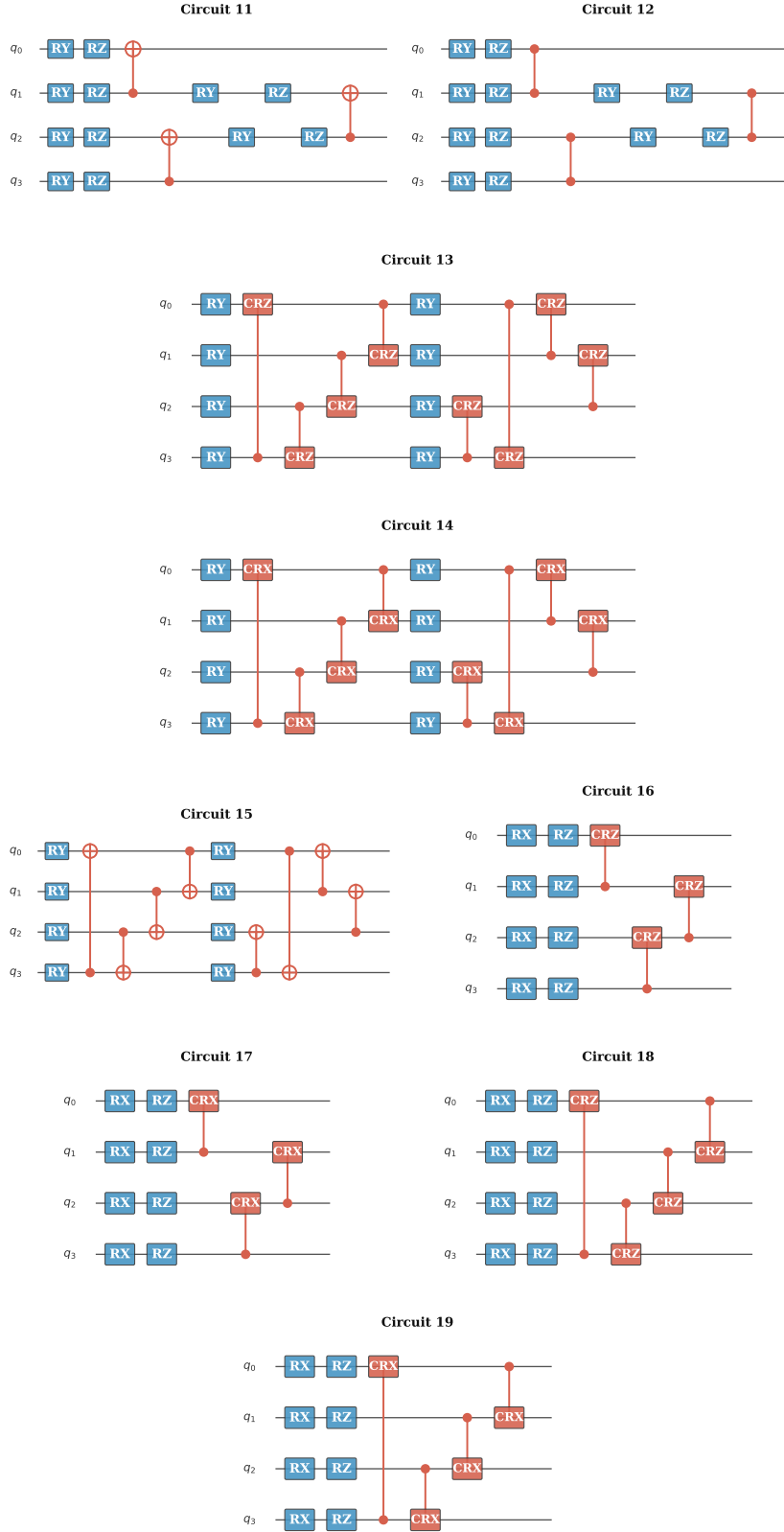


Figure 4.3: Circuit templates 11–19 from Sim et al. (2019), adapted as quantum feature maps. Blue boxes indicate single-qubit rotation gates; red elements indicate entangling gate. All circuits are shown for a single structural layer ($L = 1$) and four qubits. Encoding R_Y blocks are not shown. Source: adapted from Sim et al. (2019)

(Życzkowski & Sommers, 2005). Intuitively, a more expressible circuit is one whose output states cover the Hilbert space more evenly as its parameters vary. Formally, it is quantified as the Kullback-Leibler divergence between the distribution of state fidelities generated by the circuit and that of Haar-random states:

$$\text{Expr} = D_{\text{KL}}\left(\hat{P}_{\text{PQC}}(F) \parallel P_{\text{Haar}}(F)\right)$$

where $F = |\langle\psi_\theta|\psi_\phi\rangle|^2$ is the fidelity between two states sampled from the circuit, and $P_{\text{Haar}}(F) = (N-1)(1-F)^{N-2}$ is the analytical fidelity distribution for Haar-random states in a Hilbert space of dimension $N = 2^n$ (Sim et al., 2019). A lower D_{KL} value indicates higher expressibility.

Entangling capability measures the average amount of entanglement a circuit can generate across its parameters. It is estimated as the mean Meyer-Wallach measure Q (Meyer & Wallach, 2002) over a set of randomly sampled parameter vectors:

$$\text{Ent} = \frac{1}{|S|} \sum_{\theta_i \in S} Q(|\psi_{\theta_i}\rangle)$$

The Meyer-Wallach measure $Q \in [0, 1]$ is zero for product states and one for maximally entangled states (Sim et al., 2019). A higher Ent value indicates that the circuit tends to produce more entangled states on average.

4.5.4 Rank correlation analysis

To assess whether the structural rankings predict classification performance, Spearman's rank correlation coefficient is computed between the expressibility and entangling capability rankings reported by Sim et al. (2019) and the ranking induced by the mean test accuracy obtained experimentally. The analysis is performed separately for each dataset, as well as for an aggregate ranking obtained by averaging the mean test accuracy across all four datasets before ranking. Statistical significance is assessed using the associated p-value, with $p < 0.05$ considered significant.

4.6 Experiment 2 — Rotation axis and entanglement ablation

This experiment investigates which specific elements of a quantum circuit architecture determine its performance as a kernel. The analysis is structured as a controlled ablation

study, isolating the effect of the rotation axis and the presence of entanglement through two independent blocks of experiments. The following sections describe the circuit designs evaluated, the experimental protocol, and the metrics used for comparison.

4.6.1 Motivation

As shown in Section 4.5, the expressibility and entangling capability rankings of Sim et al. (2019) do not consistently predict classification accuracy when those circuits are used as quantum feature maps. This result, however, leaves an important question open: the nineteen circuits in that benchmark vary simultaneously in gate types, entanglement topology, circuit depth, and connectivity, making it impossible to attribute differences in accuracy to any single architectural element. A lack of correlation with the structural rankings could reflect that expressibility and entangling capability are the wrong metrics, or simply that the circuits are too heterogeneous for meaningful comparison.

To address this, Experiment 2 takes a different approach. Rather than evaluating a diverse set of pre-existing circuits, it constructs a minimal and controlled set of architectures that isolates two specific variables: the choice of rotation axis used for encoding and structural gates, and the presence and degree of entanglement. By keeping everything else fixed, any observed differences in accuracy can be attributed directly to the variable under study.

4.6.2 Circuit designs

The experiment is divided into two blocks. Block 1 isolates the effect of the rotation axis, keeping the circuit structure minimal and fixed. Block 2 isolates the effect of entanglement, keeping the encoding fixed and varying only the entangling structure. In both blocks, the same encoding convention from Section 4.1 applies: qubits q_0 and q_2 encode the first feature x_1 , and qubits q_1 and q_3 encode the second feature x_2 .

Block 1 — Rotation axis ablation

Block 1 contains twelve circuits grouped into two types. The first type consists of three encoding-only circuits, one per rotation axis (R_X , R_Y , R_Z), shown in Figure 4.4. Each circuit applies a single rotation gate per qubit, with the input features as rotation angles. These circuits have no structural parameters and serve as a minimal baseline: any

classification performance they achieve comes entirely from the geometry induced by the encoding axis, with no additional degrees of freedom.

The second type consists of nine circuits that add a single structural gate between two encoding layers, as shown in Figure 4.5. All three axes are considered for both the encoding gate and the structural gate, producing the nine possible combinations. The structural gate angle is sampled randomly and kept fixed during kernel evaluation. As discussed in Section 4.1, the circuit must end with an encoding gate to ensure the structural gate contributes to the kernel; all circuits therefore follow the structure $R_{\text{enc}}(x) \rightarrow R_{\text{param}}(\theta) \rightarrow R_{\text{enc}}(x)$.

Block 2 — Entanglement ablation

Block 2 contains three circuits that share the same encoding gate (R_Y) and vary only in the degree of entanglement, as shown in Figure 4.6. The choice of R_Y as the encoding gate is motivated by the results of Block 1 (Section 5.3): R_Z -based circuits consistently fail to learn, as phase-only rotations do not produce distinguishable kernel values under fidelity measurement (indeed, for a single qubit, $\langle 0 | R_Z(-x') R_Z(x) | 0 \rangle = 1$, so the kernel is constant). Single-qubit R_Y rotations are widely employed for data encoding in quantum machine learning architectures, making them a natural baseline for evaluating alternative encoding choices.

The three circuits are: no entanglement ($R_Y(x) \rightarrow R_Y(x)$, two encoding layers with no structural gate), partial entanglement ($R_Y(x) \rightarrow \text{CRZ}_{0 \rightarrow 1} \rightarrow \text{CRZ}_{2 \rightarrow 3} \rightarrow R_Y(x)$, two independent controlled rotation gates acting on separate qubit pairs), and full entanglement ($R_Y(x) \rightarrow \text{CRZ ring} \rightarrow R_Y(x)$, a ring of controlled rotations connecting all four qubits). The CRZ gate was chosen as the entangling operation because it introduces tunable correlations between qubits while remaining simple enough to isolate the effect of entanglement without introducing additional single-qubit degrees of freedom.

4.6.3 Experimental design

All fifteen circuits are evaluated on the four datasets described in Section 4.2, following the same protocol as Experiment 1. For circuits with structural parameters (the nine axis combinations in Block 1 and the two entangled circuits in Block 2), twenty random seeds are used to sample the structural gate angles, and mean and standard deviation of train and test accuracy are reported across seeds. Encoding-only circuits have no structural

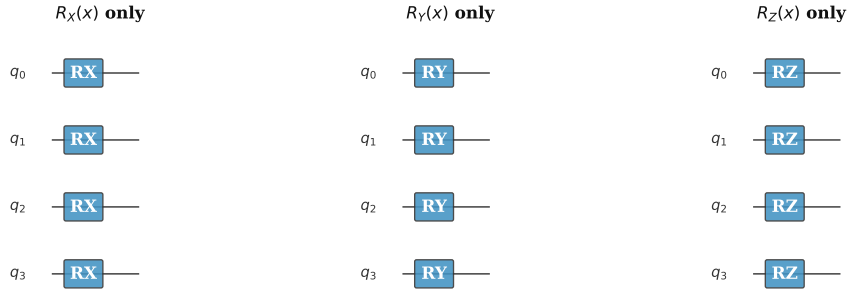


Figure 4.4: Block 1 encoding-only circuits. Each circuit applies a single rotation gate per qubit using the input features as angles. No structural parameters are present. Source: own work

parameters and are therefore evaluated with a single run per dataset.

4.7 Experiment 3 — Qubit scaling

Experiments 1 and 2 evaluate circuits with a fixed number of qubits. A natural question that follows is whether scaling the system size produces meaningful differences in classification performance. As the number of qubits n increases, the dimension of the Hilbert space grows as 2^n , which in principle allows the quantum feature map to embed data into increasingly rich spaces. Whether this translates into better kernels — or instead leads to concentration effects that degrade performance — is precisely what this experiment investigates. The three circuits selected from Experiment 2 are evaluated across four qubit counts, keeping all other conditions fixed.

4.7.1 Motivation

Experiment 2 isolates the effect of gate type and entanglement structure on classification performance, keeping the number of qubits fixed at four. There is, however, another architectural dimension that directly affects the structure of the quantum feature space: the number of qubits. With four qubits, classical data is embedded into a $2^4 = 16$ -dimensional Hilbert space. Increasing the qubit count expands this space exponentially, which could in principle allow the kernel to capture finer distinctions between data points.

A natural hypothesis is that a richer Hilbert space leads to better classification performance. However, as discussed in Section 2.3.3, this scaling also increases the risk of

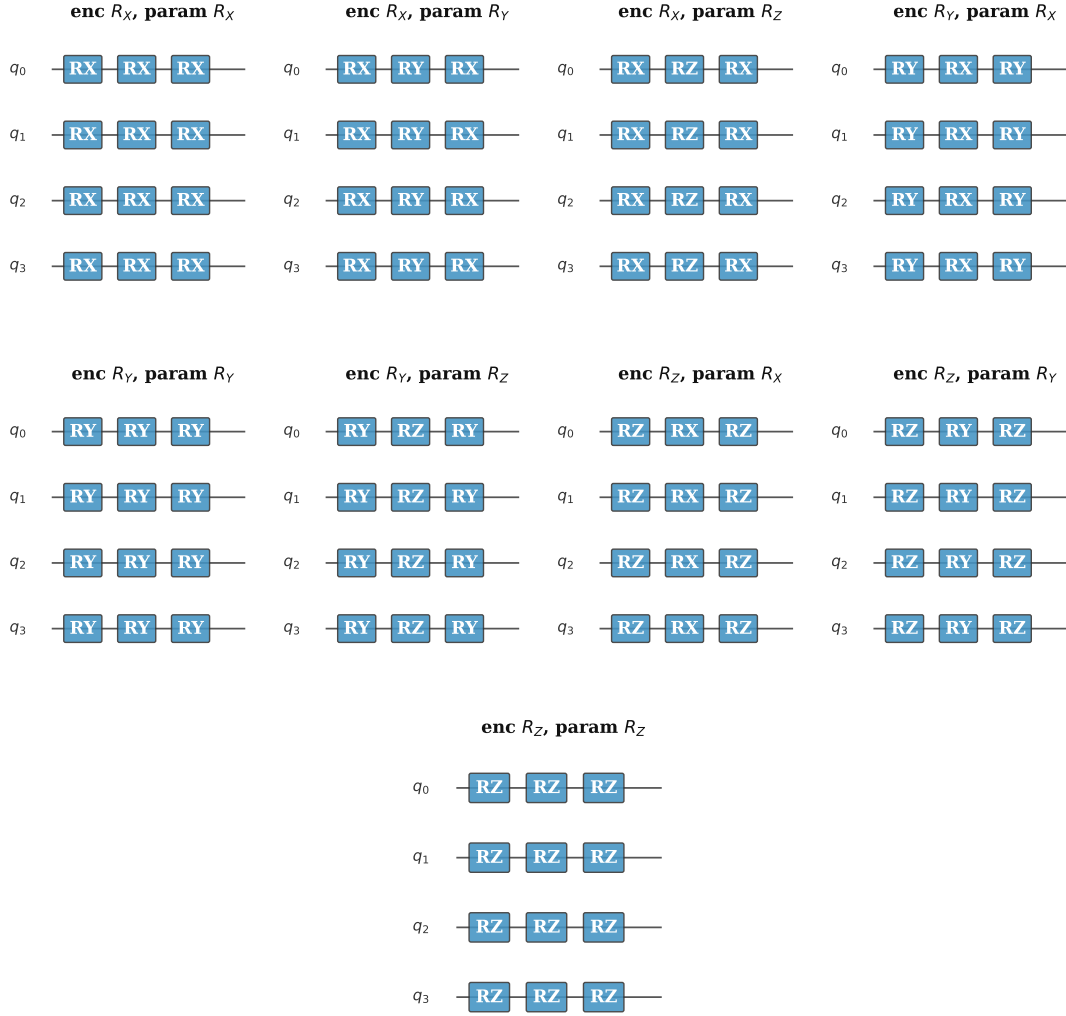


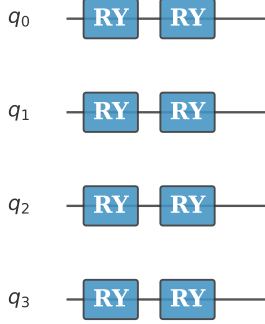
Figure 4.5: Block 1 axis combination circuits. Each circuit applies an encoding gate, a structural gate with a random angle θ , and a second encoding gate. All nine combinations of encoding and structural axes are evaluated. Source: own work.

exponential concentration: a phenomenon in which kernel values collapse toward a constant as the system grows, making the kernel unable to distinguish between inputs. This experiment investigates the role of qubit count by fixing all other circuit properties and varying only the system size, using the three circuits selected from Experiment 2 as representatives of each architectural family.

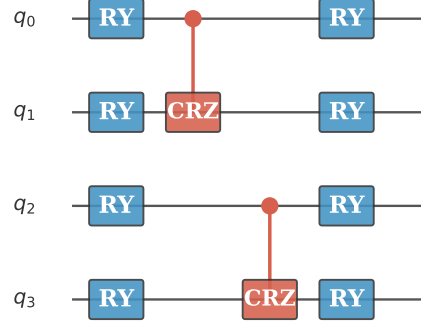
4.7.2 Circuit selection and scaling design

To isolate the effect of qubit count, three circuits are selected from Experiment 2, one per architectural group: the best-performing circuit from the encoding-only group ($R_X(x)$),

RY — no entanglement



RY — partial entanglement



RY — full entanglement (ring)

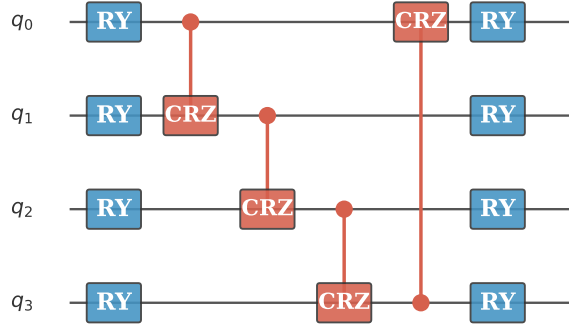


Figure 4.6: Block 2 entanglement ablation circuits. All three circuits use R_Y encoding. Source: own work.

from the encoding-structural group ($R_Y(x) \rightarrow R_Z(\theta) \rightarrow R_Y(x)$), and from the entanglement group (R_Y with partial entanglement). Each circuit is then evaluated at four qubit counts: 2, 4, 8, and 16. Powers of two were selected to examine how performance evolves under successive doublings of the Hilbert space dimension while keeping the scaling pattern simple and comparable across experiments.

With two qubits, each input feature is assigned to exactly one qubit: q_0 encodes x_1 and q_1 encodes x_2 . For larger qubit counts, the same redundant encoding strategy used in previous experiments is applied: qubit q_i encodes feature $x_{(i \bmod 2)+1}$, so that each input dimension is encoded by $n/2$ qubits in a circuit of width n . For the partial entanglement circuit, the entangling structure scales accordingly: independent CRZ pairs are applied to qubits $(0, 1), (2, 3), (4, 5), \dots$, maintaining the same local connectivity pattern as the original four-qubit design.

4.7.3 Experimental design

The experimental protocol follows the same design as in previous experiments. All circuits with structural parameters are evaluated across twenty random seeds, and mean and standard deviation of train and test accuracy are reported across seeds and datasets. The encoding-only circuit ($R_X(x)$) has no structural parameters and is therefore evaluated with a single run per dataset. The four datasets described in Section 4.2 are used throughout.

4.8 Experiment 4 - Exponential concentration and embedding analysis

This experiment investigates the mechanisms underlying the performance trends observed in the previous experiments. Rather than focusing on classification accuracy alone, it analyzes the geometry of the kernel matrix and the corresponding quantum feature maps to determine how circuit scaling affects the quality of the resulting representations.

4.8.1 Motivation

The results of Experiment 3 suggest that increasing the number of qubits does not consistently improve classification performance. As discussed in Section 5.4, some circuits show a degradation in test accuracy beyond a certain system size, which contradicts the intuition that a larger Hilbert space should always produce a more expressive kernel. This pattern is consistent with a known phenomenon in quantum kernel methods called exponential concentration: as the number of qubits grows, kernel values between different data points tend to collapse towards a common constant, making the kernel matrix nearly uniform and the classifier unable to distinguish between inputs. The concentration point for the fidelity kernel is $1/2^n$, which decreases exponentially with the number of qubits n .

What drives this phenomenon, and under which conditions it emerges, is less clear. Thanasilp et al. (2024) identify several potential sources — including the expressivity of the embedding, the presence of entanglement, circuit depth, and system size — but the relative contribution of each factor, and how they interact, remains an open empirical question for concrete circuit architectures. Experiment 4 investigates this by evaluating three variants of the same circuit across four qubit counts and four layer counts, varying entanglement structure, depth, and system size in a controlled way. The goal is to characterize how and

when concentration emerges in practice, and which of these factors plays a more prominent role.

4.8.2 Circuit architecture

The circuits used in the previous experiments were designed with a specific purpose: isolating the effect of individual architectural choices such as rotation axis or entanglement degree. While this controlled design is appropriate for those experiments, studying exponential concentration calls for a different approach. Concentration is a general phenomenon that affects quantum kernels broadly, and characterizing it on circuits tailored to the experimental pipeline of this Work would limit the generality of the conclusions. A circuit that happens to perform well or poorly for reasons unrelated to concentration could obscure the effect under study.

For this reason, Experiment 4 uses the ZZ feature map introduced by Havlíček et al. (2019), one of the most widely studied quantum feature maps in the literature (Schuld, 2021). Using an established circuit offers a concrete advantage: any concentration effects observed can be attributed to the architectural properties under study — entanglement, depth, and system size — rather than to arbitrary design choices. This grounding in a well-understood encoding strengthens the interpretability of the results.

The ZZ feature map encodes classical data through repeated applications of the following layer structure. Each layer begins with a Hadamard gate applied to every qubit, followed by single-qubit R_Z rotations whose angles are proportional to the input features:

$$R_Z(2x_i), \quad i = 1, \dots, n$$

This is then followed by a block of two-qubit interactions. For each pair of qubits (i, j) in the entanglement topology, the interaction is implemented as a CNOT– R_Z –CNOT sequence, where the rotation angle encodes a second-order product of the input features:

$$R_Z(2(\pi - x_i)(\pi - x_j))$$

This structure introduces data-dependent correlations between qubits that go beyond the single-qubit encoding used in Experiments 1–3, and is the key feature that gives the ZZ feature map its expressibility. The full circuit is obtained by repeating this layer L times.

It is worth noting that for the first layer, since the Hadamard gates act on the initial state $|0\rangle^{\otimes n}$, the resulting computational-basis probabilities are uniform regardless of the

input data or the entanglement structure — both R_Z rotations and the CNOT– R_Z –CNOT interaction only modify the relative phase between basis states, leaving their magnitude unchanged. From the second layer onward, the Hadamard gates act on a state that already carries data-dependent phase information from the previous layer, so this uniformity no longer holds. This distinction is relevant when interpreting embedding-level metrics, discussed further in Chapter 5.

Three variants of the entanglement topology are evaluated, shown in Figure 4.7 for a representative case of $n = 4$ qubits and $L = 1$ layer. The first uses no entanglement, reducing the circuit to a sequence of Hadamard and R_Z gates with no two-qubit interactions. The second applies entangling gates between nearest-neighbor pairs only, following a linear topology. The third connects all pairs of qubits, producing a fully entangled embedding. Each of the three variants is evaluated across four qubit counts ($n \in \{4, 8, 12, 16\}$) and four layer counts ($L \in \{1, 2, 4, 8\}$), yielding a grid of 48 configurations per dataset. Together, this design allows the individual contributions of entanglement degree, circuit depth, and system size to be examined within a single coherent framework.

4.8.3 Experimental design

The experimental grid covers all combinations of three circuit variants, four qubit counts ($n \in \{4, 8, 12, 16\}$), four layer counts ($L \in \{1, 2, 4, 8\}$), and the four datasets described in Section 4.2, yielding 192 configurations in total. Unlike the previous experiments, the ZZ feature map contains no random structural parameters. Consequently, running multiple seeds for circuit initialization would yield identical kernel matrices. To obtain a meaningful estimate of the classifier's performance and avoid overfitting to a particular data split, the experimental design adopts a different strategy: the train-test split is varied across several random seeds (instead of the circuit parameters). This approach isolates the variability due to data subsampling, providing a more robust evaluation of the kernel's generalization capacity.

The analysis is organized into two complementary phases. The first focuses on kernel-level metrics: for each configuration, the quantum kernel matrix is computed and used to train an SVM classifier, from which classification performance and a set of kernel quality metrics are extracted. These metrics, defined in Chapter 2, characterize the degree of concentration directly from the structure of the kernel matrix and the behavior of the classifier. The second phase focuses on the geometry of the quantum embedding itself:

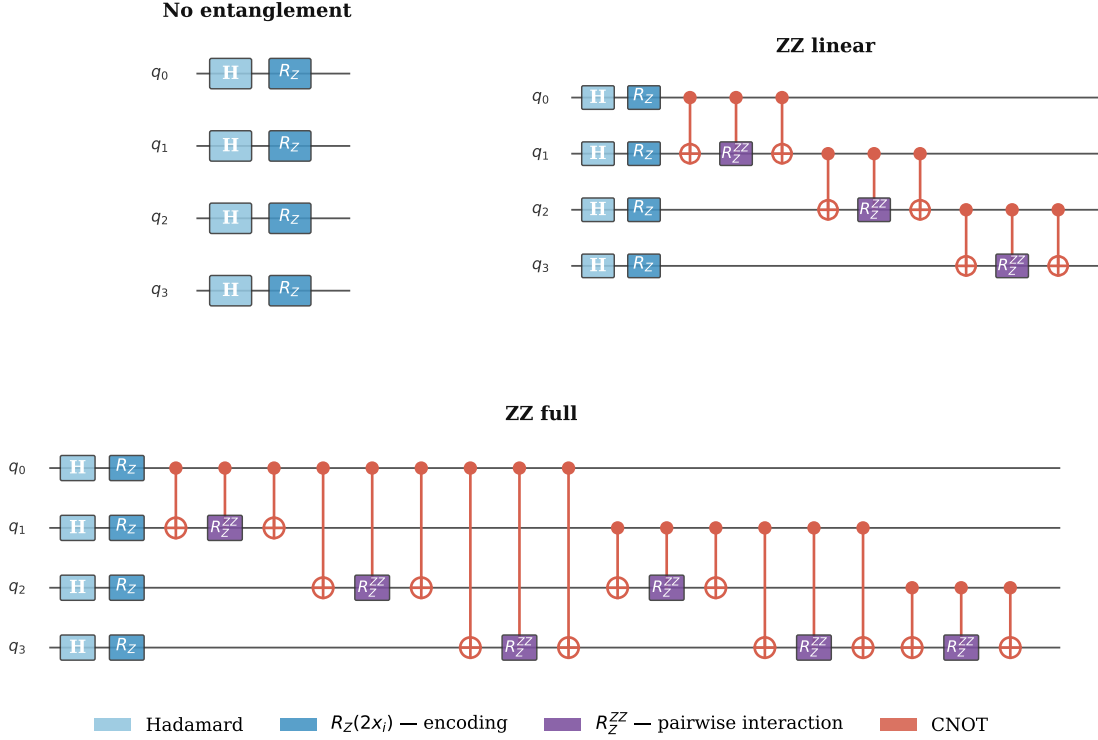


Figure 4.7: The three ZZ feature map variants evaluated in Experiment 4, shown for $n = 4$ qubits and $L = 1$ layer. Light blue boxes indicate Hadamard gates, blue boxes indicate $R_z(2x_i)$ encoding rotations, purple boxes indicate the pairwise R_{ZZ} interaction gates, and red elements indicate CNOT gates. From left to right: no entanglement, linear entanglement, and full entanglement. Source: own work.

rather than looking at the kernel matrix, it examines how individual data points are distributed across the Hilbert space after encoding, providing a complementary view of what happens inside the feature map as the system scales. Both phases are run independently and their results are analyzed jointly in Chapter 5.

In addition, SVM decision boundaries are once again plotted over the input space to provide a visual illustration of how the classifier degrades as concentration increases. Decision boundary visualization requires evaluating the classifier over a dense grid of points, and this cost scales with circuit depth and entanglement. For the entangled topologies, generating boundaries at $L = 4$ with sufficient grid resolution was computationally expensive on the available cluster resources. Boundaries are therefore shown at $L = 2$, which is sufficient to illustrate the qualitative separability pattern of each topology.

5. Results

This Chapter presents the results obtained in the four experiments introduced in Chapter 4. The analyses progress from overall classification performance to a more detailed study of the quantum kernels and the corresponding embedding structure, allowing the observed trends to be interpreted from both a predictive and a geometric perspective.

5.1 Classical kernels

Table 5.1: Test accuracy of classical SVM kernels across datasets after hyperparameter optimization. Source: own work.

Dataset	Linear	Poly	RBF
Checkerboard	0.542	0.650	0.958
Circles	0.583	0.992	0.992
Moons	0.908	0.917	1.000
XOR	0.700	1.000	1.000

The results in Table 5.1 can be interpreted as a classical benchmark for dataset separability under increasing levels of model flexibility. Rather than focusing only on the best-performing kernel, the relevant aspect is how performance changes from linear to polynomial and finally to RBF models.

The linear kernel provides a direct indication of how close each dataset is to being linearly separable. In this sense, Checkerboard exhibits the strongest departure from linear separability, whereas Moons can already be classified reasonably well with a linear decision boundary. By contrast, Circles and XOR benefit substantially from nonlinear feature mappings.

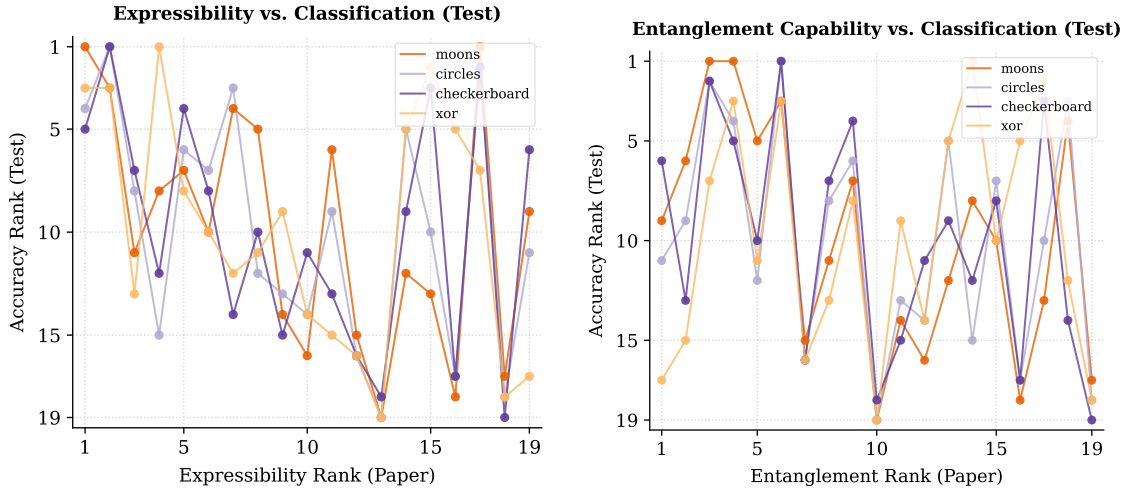
Polynomial kernels substantially improve performance over the linear model for the Circles and XOR datasets, where they nearly reach perfect classification. Checkerboard also benefits from the nonlinear mapping, although the improvement is considerably smaller and the resulting accuracy remains well below that achieved by the RBF kernel.

Finally, the RBF kernel provides a strong nonlinear baseline that performs well across all datasets. While it does not distinguish as clearly between datasets in terms of raw

accuracy, its role is mainly to indicate the upper bound of classical kernel performance. Small differences in this regime are still relevant, as they reflect how close each dataset is to being fully captured by standard kernel methods.

Overall, this benchmark provides a reference for classical separability that is later used to contextualize the performance of quantum kernels, both in terms of achievable accuracy and in terms of how much additional structure is required beyond standard nonlinear mappings.

5.2 Experiment 1 — Circuit architecture as a predictor of kernel performance



(a) Expressibility ranking vs. test accuracy. (b) Entangling capability ranking vs. test accuracy.

Figure 5.1: Correlation between the rankings proposed by Sim et al. (2019) and test accuracy. (a) Expressibility ranking. (b) Entangling capability ranking. Source: own work.

The results show a moderate positive relationship between the theoretical rankings proposed by Sim et al. (2019) and the classification performance obtained in the experiments. As shown in Figure 5.1 and Table 5.2, the strongest agreement is observed for the Moons dataset, where both expressibility and entangling capability present statistically significant correlations. When all datasets are considered together, only expressibility remains significantly correlated with the test accuracy.

Table 5.2: Spearman rank correlation between the theoretical rankings of Sim et al. (2019) and the classification performance obtained for each dataset. ρ is the correlation and p the p-value. Source: own work.

Dataset	Expressibility		Entangling capability	
	ρ	p	ρ	p
Checkerboard	0.296	0.218	0.414	0.078
Circles	0.409	0.082	0.291	0.226
Moons	0.470	0.042	0.506	0.027
XOR	0.371	0.117	-0.143	0.559
Aggregate	0.477	0.039	0.185	0.448

Overall, expressibility appears to be a more consistent indicator of classification performance than entangling capability, although the observed correlations are only moderate. At the same time, the results suggest that neither property alone is sufficient to explain the behaviour of the quantum kernels across all datasets.

A clear relationship between entangling capability and dataset complexity cannot be established from these experiments. While the Moons dataset shows the strongest positive correlation, the results for Checkerboard and Circles are weaker and not statistically significant. This indicates that the benefit of entanglement depends on factors beyond the non-linearity of the classification task.

Similarly, although expressibility exhibits a more stable correlation with performance, it does not consistently predict the behaviour of the kernels on every dataset. For instance, its correlation is relatively weak for the Checkerboard dataset despite being the most challenging classification problem considered in this Work. These observations suggest that additional circuit characteristics, beyond expressibility and entangling capability, also influence the final classification performance.

5.3 Experiment 2 — Rotation axis and entanglement ablation

The previous experiment showed that global circuit properties such as expressibility and entangling capability only partially explain the observed classification performance. To obtain a more direct understanding of the role played by different architectural choices, this experiment analyses the contribution of the main building blocks of the quantum feature maps by evaluating the effect of the encoding gate, the addition of parametrized single-qubit rotations, and the inclusion of entangling operations.

As shown in Figure 5.2, using only an encoding gate is already sufficient to achieve high classification accuracy on the Moons, Circles and XOR datasets when the data are encoded through R_X or R_Y . In contrast, circuits based exclusively on R_Z encoding perform considerably worse. This behaviour is expected, since rotations around the Z axis only modify the phase of the quantum state and, without additional operations, these phase differences cannot be translated into observable differences in the measurement outcomes.

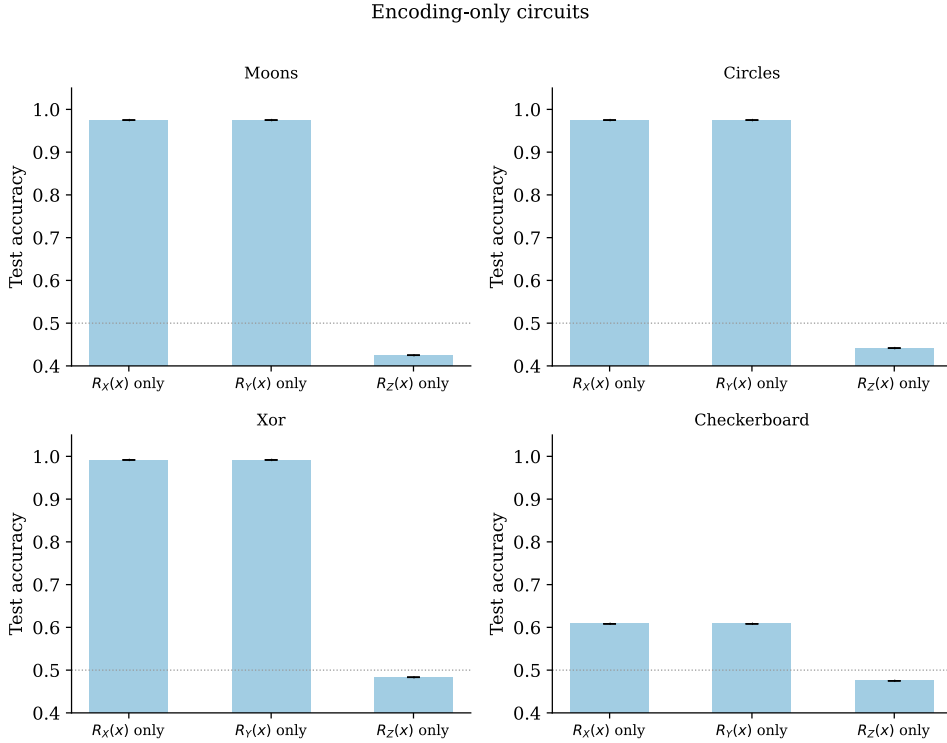


Figure 5.2: Classification performance of the encoding-only circuits. Source: own work.

The effect of introducing parametrized single-qubit rotations is shown in Figure 5.3.

Overall, these circuits outperform the encoding-only configurations. In particular, combining R_Z encoding with rotations around the X or Y axes largely removes the limitations observed previously. However, despite this improvement, all rotation-based configurations still perform relatively poorly on the Checkerboard dataset, suggesting that single-qubit operations alone are insufficient for the most demanding classification task considered in this Work.

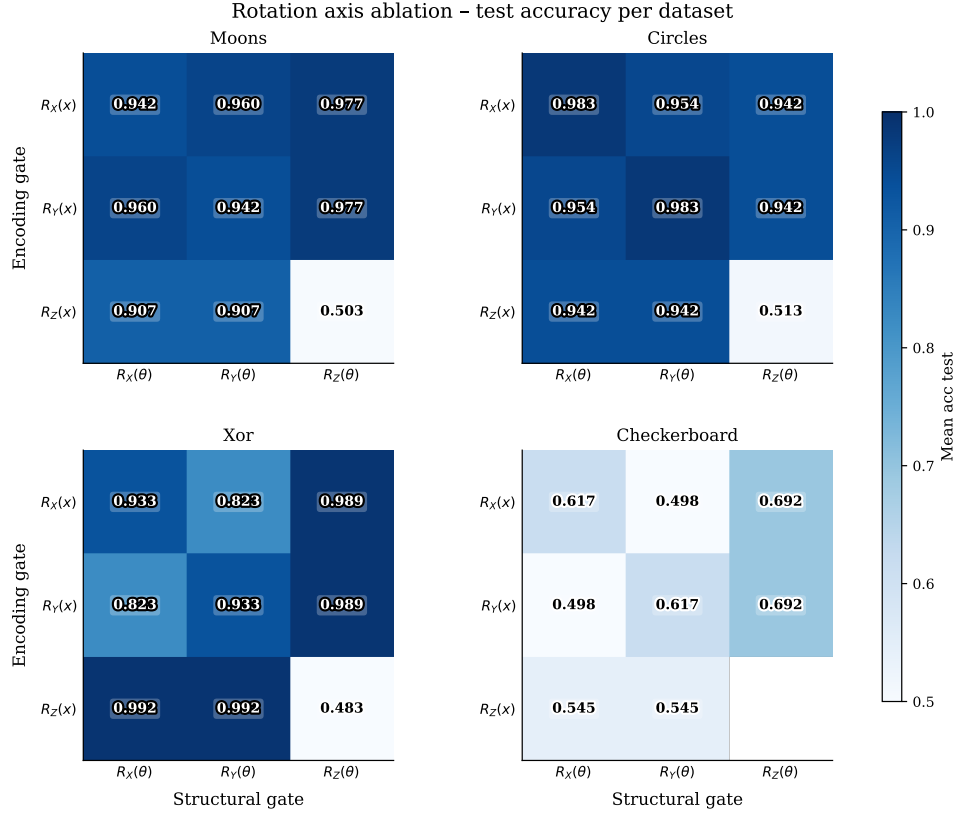


Figure 5.3: Classification performance of the rotation circuits. Source: own work.

The impact of entanglement is presented in Figure 5.4. Unlike the previous circuit blocks, the introduction of entangling gates produces the largest improvements on the Moons and, especially, the Checkerboard datasets. This observation is consistent with the findings from Experiment 1, where entanglement was also found to be most beneficial for these datasets. Interestingly, the partial entanglement configuration slightly outperforms the full configuration in both cases.

It should be noted that the partial circuit simply contains fewer entangling gates than the full configuration. This does not necessarily imply that the resulting quantum states are less entangled, since the amount of entanglement generated depends on both the circuit

structure and the encoded data. Therefore, the observed differences should be interpreted as the effect of the circuit architecture rather than as a direct consequence of a higher or lower degree of quantum entanglement.

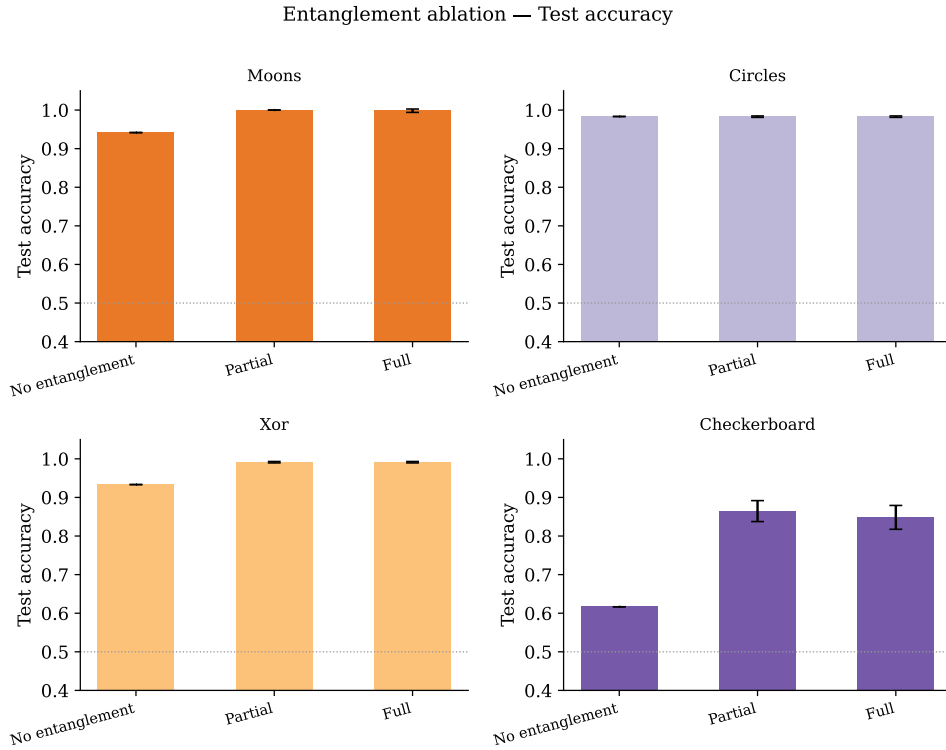


Figure 5.4: Classification performance of the entanglement circuits. Source: own work.

Overall, these observations indicate that the choice of encoding axis has the greatest impact on the simpler datasets, whereas entanglement becomes increasingly important for more challenging classification tasks. At the same time, increasing the number of entangling gates beyond a certain point does not necessarily translate into better classification performance. The complete numerical results for all circuit configurations are provided in Appendix A.1.

5.4 Experiment 3 — Qubit scaling

Having identified the best-performing configuration within each circuit family, the next step is to analyse how these representative architectures scale as the size of the quantum system increases. An overview of the behaviour of the three selected circuits is provided in Figure 5.5.

Overall, increasing the number of qubits improves the performance of all three archi-

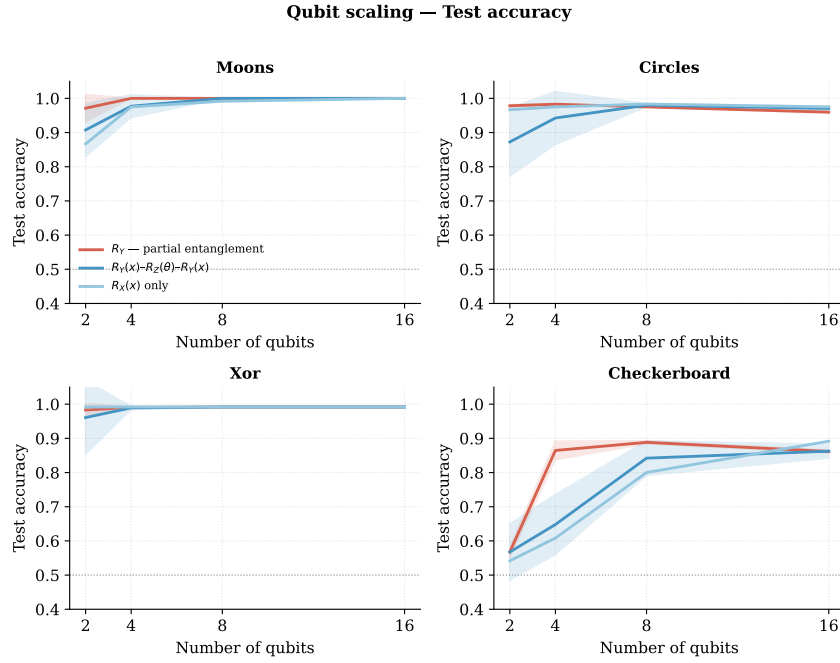


Figure 5.5: Classification performance of all circuits with growing amount of qubits. Source: own work.

tectures, although the magnitude of this improvement depends on the circuit design. The encoding-only circuit exhibits the largest gains as the system size increases, particularly on the Checkerboard dataset, where competitive accuracy is only achieved with a larger number of qubits. In contrast, the rotation circuit follows a similar trend but reaches a plateau earlier, with only marginal improvements beyond 8 qubits. On simpler datasets, such as Moons and XOR, both architectures rapidly approach their maximum performance, indicating that additional qubits provide little benefit once the decision boundary has become sufficiently expressive. Interestingly, on Circles, the encoding-only circuit performs slightly better with 8 qubits than with 16, suggesting that increasing the number of qubits does not necessarily improve generalization indefinitely.

As the number of qubits increases, the influence of the random structural parameters becomes progressively weaker. This trend is particularly evident for the entanglement circuit, whose standard deviation rapidly decreases as the system grows, indicating that different random initializations lead to increasingly similar classification performance. In contrast, the rotation circuit exhibits noticeably larger variability, and this variability persists over a wider range of qubit counts before eventually diminishing. The encoding-only circuit, having no random structural parameters, is deterministic and therefore shows

no variability across runs.

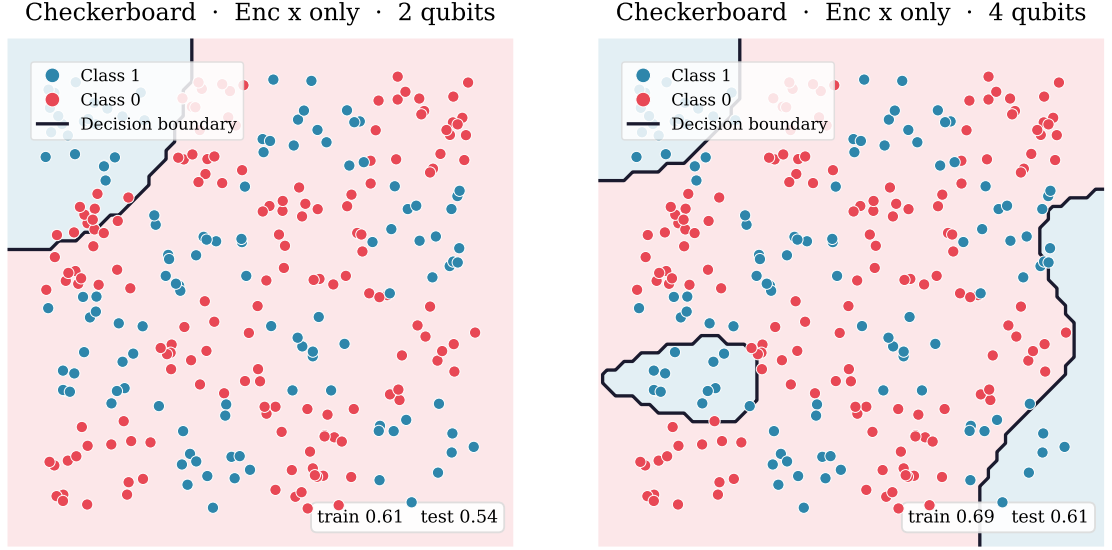
The qualitative effect of increasing the system size is illustrated in Figure 5.6. With only 2 qubits, the decision boundary captures almost none of the checkerboard structure, correctly identifying only a small region of the input space. As the number of qubits increases, additional regions gradually appear and the boundary becomes increasingly structured. By 16 qubits, the classifier reproduces most of the alternating pattern, illustrating how larger quantum systems can generate considerably richer decision surfaces even without introducing additional circuit components.

The scaling behaviour differs for the entangling architecture. As shown in Figure 5.5, increasing the number of qubits initially improves performance, but the gains become progressively smaller than for the previous architectures. A more detailed view is provided in Figure 5.7, which includes both training and test accuracy. While the training accuracy continues to improve as the number of qubits increases, the test accuracy reaches a maximum and then slightly decreases, particularly on the Circles and Checkerboard datasets. This behaviour is consistent with the onset of overfitting and suggests that increasing the system size alone is not always beneficial once the circuit already possesses a sufficiently expressive architecture. Although the present experiment does not explain the origin of this behaviour, it indicates that larger quantum systems may progressively lose their ability to generalize. The underlying causes are investigated in the following experiment through an analysis of the resulting kernel matrices and quantum embeddings.

Overall, these results indicate that the benefits of increasing the number of qubits depend strongly on the circuit architecture. Simpler circuits continue to benefit from larger quantum systems because their expressive power remains limited, whereas more expressive architectures reach a point where further scaling provides little or no improvement in generalization. Complete decision boundaries for the remaining datasets are provided in Appendix A.2.

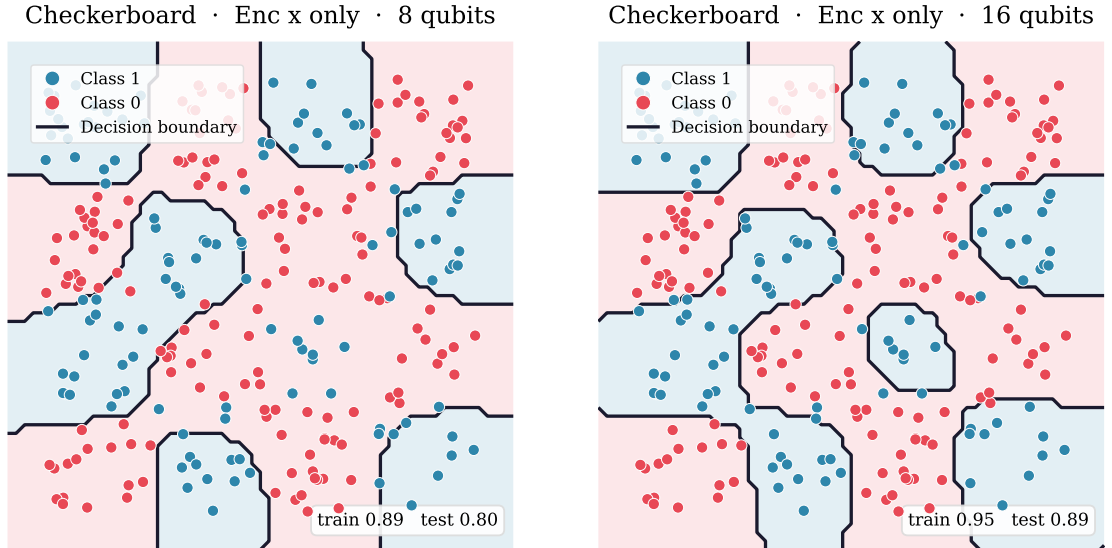
5.5 Experiment 4 - Exponential concentration and embedding analysis

The previous experiments focused on relatively simple circuit families and showed that increasing circuit complexity does not necessarily improve classification performance. This final experiment therefore considers more expressive ZZ feature maps in order to investigate



(a) Decision boundaries of the encoding-only circuit for 2 qubits on Checkerboard

(b) Decision boundaries of the encoding-only circuit for 4 qubits on Checkerboard



(c) Decision boundaries of the encoding-only circuit for 8 qubits on Checkerboard

(d) Decision boundaries of the encoding-only circuit for 16 qubits on Checkerboard

Figure 5.6: Decision boundaries of the encoding-only circuit on Checkerboard. Source: own work.

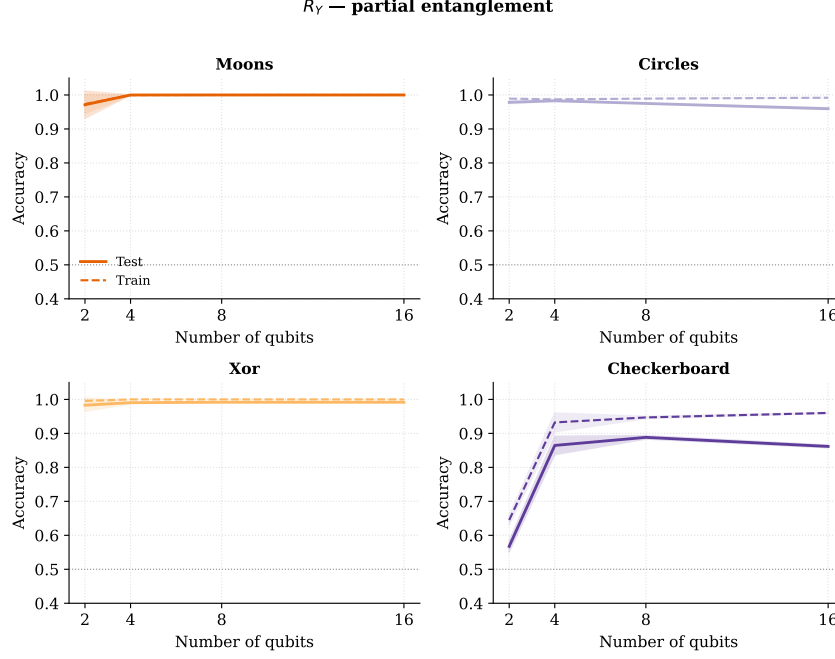


Figure 5.7: Classification performance of the entanglement circuit with growing amount of qubits. Source: own work.

the origin of this behaviour from the perspective of kernel concentration and embedding geometry.

5.5.1 Kernel analysis

The previous experiments showed that increasing the number of qubits and the circuit complexity does not necessarily improve classification performance. To better understand this behaviour, this section analyses how the properties of the quantum kernel evolve as the circuits become larger and more entangled. The figures shown correspond to circuits with four layers. The remaining layer configurations exhibit the same qualitative behaviour and are included in Appendix A.3.1.

The first indication of kernel saturation can be observed in Figure 5.8. Unlike the simpler circuit families analysed in the previous experiments, increasing the number of qubits does not lead to uniform improvements across the three topologies. The non-entangling circuit generally continues to improve with system size, whereas the gains become progressively smaller for the linear-entanglement topology and largely disappear for the fully entangling circuit, where test accuracy often stagnates or even decreases. This behaviour becomes increasingly evident as the amount of entanglement grows.

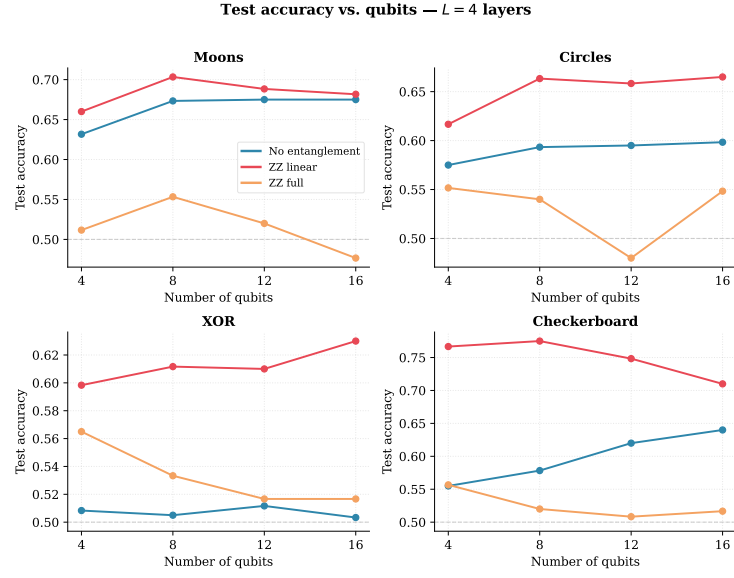


Figure 5.8: Test accuracy as a function of the number of qubits for the three circuit topologies with four layers. Source: own work.

The train–test gap provides further insight into this behaviour. As shown in Figure 5.9, training accuracy continues to improve with both the number of qubits and the amount of entanglement, while test accuracy remains approximately constant or deteriorates. Consequently, the gap between training and test performance widens, indicating progressively stronger overfitting. The complete numerical results are reported in Appendix A.

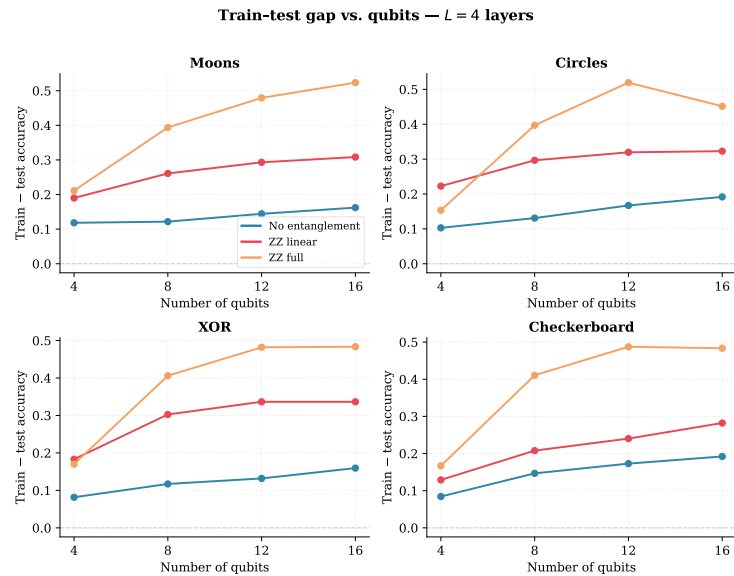


Figure 5.9: Train–test accuracy gap as a function of the number of qubits for the three circuit topologies with four layers. Source: own work.

To understand the origin of this behaviour, the structure of the kernel itself must be examined. Figure 5.10 shows that the off-diagonal variance decreases steadily as both the number of qubits and the amount of entanglement increase. This indicates that the similarities between different data points become progressively more uniform, reducing the amount of discriminative information contained in the kernel.

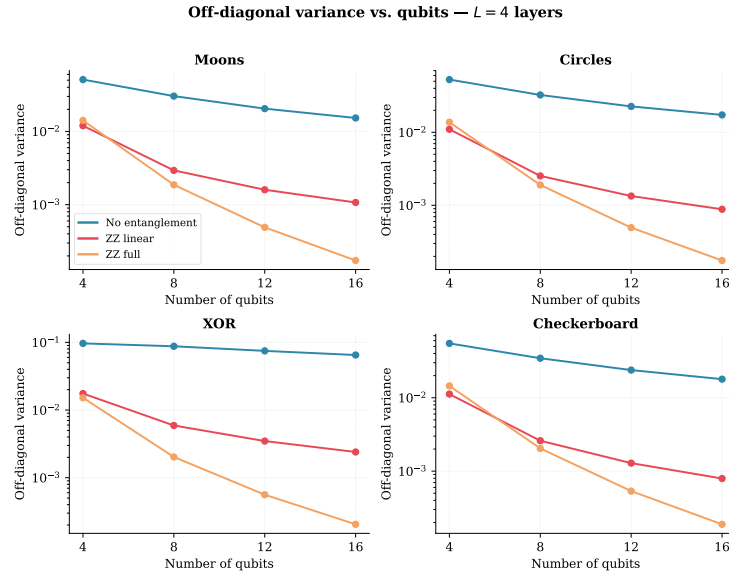


Figure 5.10: Off-diagonal variance of the kernel matrix as a function of the number of qubits for the three circuit topologies with four layers. Source: own work.

A complementary perspective is provided by the effective dimension in Figure 5.11. More entangling circuits consistently produce kernels with larger effective dimensions, particularly for larger numbers of qubits. Although a higher effective dimension is often associated with richer representations, these results show that this interpretation is not always appropriate. Here, the increase occurs simultaneously with the loss of off-diagonal variance, indicating that the kernel approaches a regime in which data points become increasingly isolated from one another rather than organised according to the underlying class structure.

The practical consequences of this loss of structure are reflected in Figure 5.12. As the kernel becomes more concentrated, the fraction of support vectors increases substantially, especially for the entangling circuits. In the most extreme cases, almost every training sample is required to define the separating hyperplane, suggesting that the classifier is relying on individual examples instead of capturing a meaningful global decision boundary.

This behaviour can be observed directly in Figure 5.13. Despite using the same number

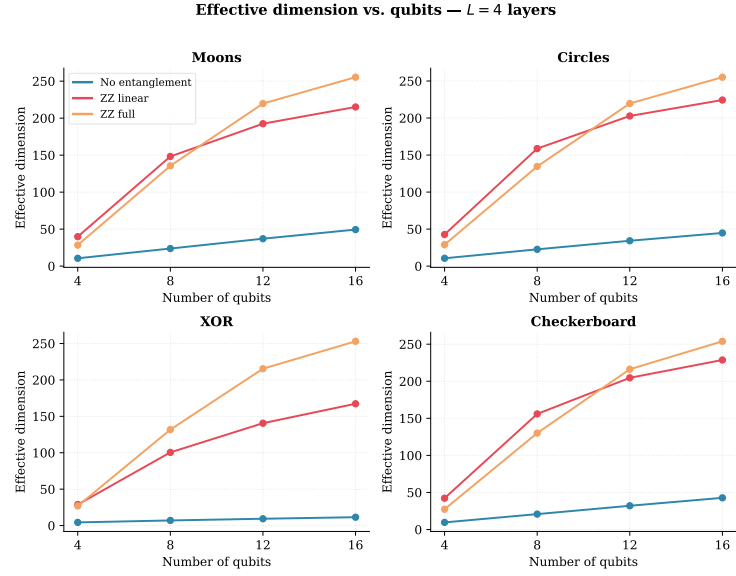


Figure 5.11: Effective dimension of the kernel matrix as a function of the number of qubits for the three circuit topologies with four layers. Source: own work.

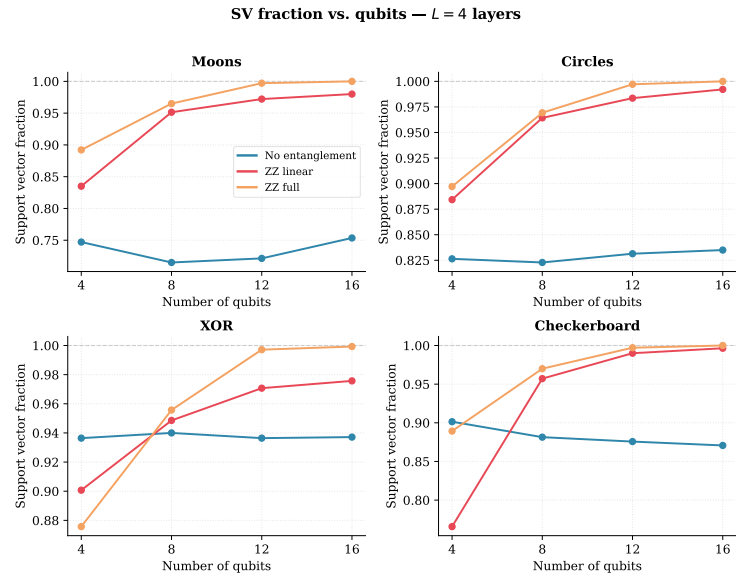
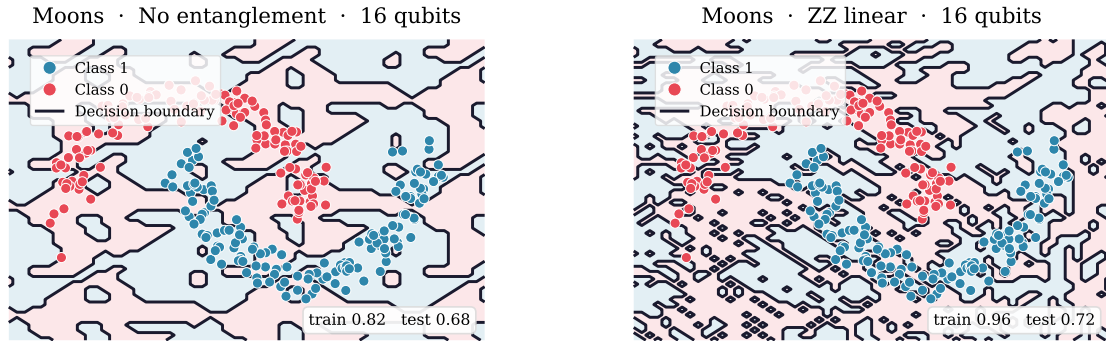


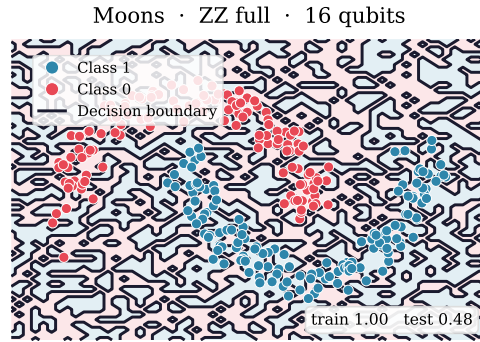
Figure 5.12: Fraction of support vectors as a function of the number of qubits for the three circuit topologies with four layers. Source: own work.

of qubits, the three circuits produce decision boundaries with noticeably different levels of fragmentation. The circuit without entanglement still preserves some coherent large-scale regions, whereas the fully entangling circuit divides the input space into many small disconnected regions. These visual differences are consistent with the increasing support vector fraction and the larger train-test gap observed previously, illustrating how the progressive loss of kernel structure ultimately translates into poorer generalization.



(a) Decision boundaries of the no-entanglement circuit for 16 qubits and $L=2$ on the Moons dataset.

(b) Decision boundaries of the linear-entanglement circuit for 16 qubits and $L=2$ on the Moons dataset.



(c) Decision boundaries of the full-entanglement circuit for 16 qubits and $L=2$ on the Moons dataset.

Figure 5.13: Decision boundaries on the Moons dataset. Source: own work.

5.5.2 Embedding analysis

The previous subsection showed that increasing circuit complexity progressively degrades the structure of the kernel. To better understand the origin of this behaviour, this sub-

section examines the corresponding quantum embeddings. All results shown here correspond to circuits with four layers. The remaining layer configurations are provided in Appendix A.3.3.

Three complementary metrics are considered. Shannon entropy measures how evenly the probability mass is distributed over the computational basis, the participation ratio estimates how many basis states contribute significantly to the embedding, and the class centroid distance evaluates the separation between the embedded classes in Hilbert space.

For a single layer, all three circuit topologies produce the maximum normalized Shannon entropy. This follows directly from the circuit structure: the initial Hadamard layer creates a uniform probability distribution, while the subsequent diagonal Z and ZZ gates only modify the phases of the amplitudes without affecting the measurement probabilities. Since this behaviour is identical for all circuit topologies, it is omitted here and discussed in Appendix A.3.3.

Figure 5.14 shows the normalized Shannon entropy for the four-layer circuits. In the non-entangling architecture, the normalized entropy remains nearly constant as the number of qubits increases. This behaviour follows from the fact that qubits evolve independently, so the probability distribution factorizes and the total entropy is additive,

$$H = \sum_i H_i.$$

Consequently, the entropy grows approximately linearly with the number of qubits and, after normalization,

$$H_{\text{norm}} = \frac{H}{n},$$

its dependence on the system size largely disappears.

With ZZ interactions, this factorization no longer holds. Consequently, no analogous conservation argument exists, and the dependence of the normalized entropy on the number of qubits reflects genuine collective effects rather than a trivial scaling with the Hilbert space dimension. The linear-entanglement topology shows a gradual increase in normalized entropy as more qubits are added, indicating that the probability distribution becomes progressively more spread across the computational basis. By contrast, the fully entangling topology initially follows a similar trend but eventually saturates and falls below the linear architecture. This behaviour is particularly noteworthy because the fully entangling circuit contains strictly more two-qubit gates than the linear topology at every system size, yet produces a less dispersed embedding. A detailed explanation of this effect is beyond the

scope of this Work, although it may be related to the much higher density of overlapping ZZ phase interactions in the fully connected topology.

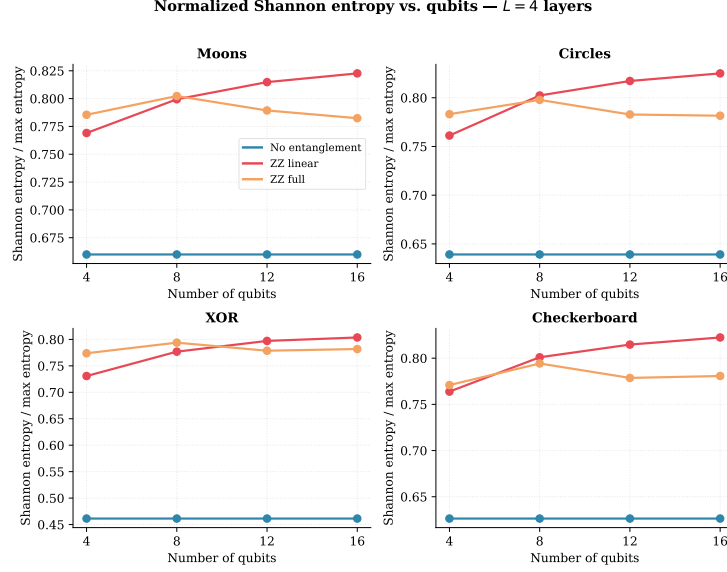


Figure 5.14: Normalized Shannon entropy of the quantum embeddings as a function of the number of qubits for the three circuit topologies with four layers. Source: own work.

The participation ratio, shown in Figure 5.15, is consistent with this picture. Across most datasets, the non-entangling and linear-entanglement circuits increase at a similar rate and remain relatively close to one another, whereas the fully entangling circuit follows a markedly different trend, growing more slowly and reaching participation ratios approximately one order of magnitude lower at 16 qubits. Despite containing the largest number of entangling gates, the fully entangling topology therefore exhibits the smallest effective support over the computational basis. The XOR dataset constitutes a partial exception, where the non-entangling circuit initially remains close to the fully entangling topology before exceeding it at larger qubit counts. Overall, the results indicate that increasing the amount of entanglement does not necessarily produce embeddings distributed over a larger number of basis states.

While entropy and participation ratio characterize how probability is distributed across basis states, they do not indicate whether the resulting embeddings preserve the separation between classes. This aspect is evaluated through the class centroid distance shown in Figure 5.16. Unlike the kernel metrics analysed in the previous subsection, the centroid distance does not exhibit a general tendency to decrease as the number of qubits increases. Instead, the non-entangling circuit shows a clear monotonic increase across all datasets,

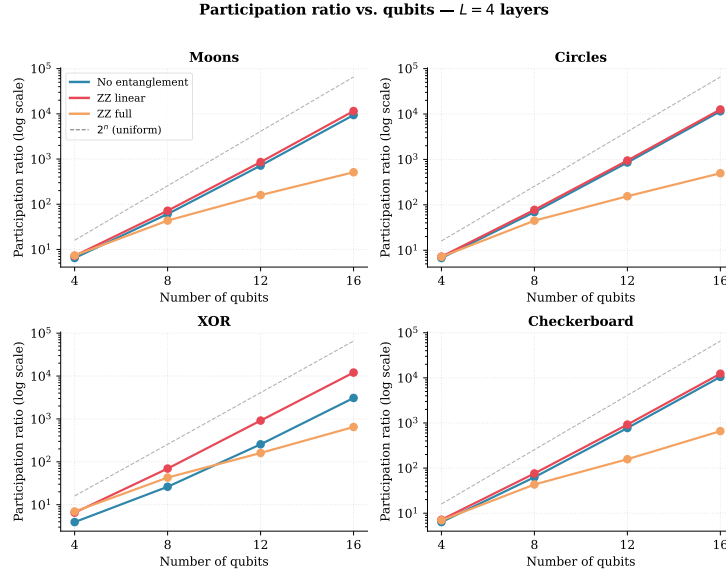


Figure 5.15: Participation ratio of the quantum embeddings as a function of the number of qubits for the three circuit topologies with four layers. Source: own work.

while the linear-entanglement topology follows a similar, although generally weaker, trend. The fully entangling circuit behaves differently, with the centroid distance remaining comparatively stable and exhibiting noticeable dataset-dependent fluctuations rather than a systematic dependence on the number of qubits. Across most datasets, however, both entangling topologies maintain smaller centroid distances than the non-entangling circuit throughout the qubit range considered, indicating that entanglement consistently reduces the absolute separation between the embedded class centroids, even if this separation does not continue to shrink as the system size grows.

Taken together, these three metrics show that increasing circuit complexity does not simply produce richer quantum embeddings. Although the fully entangling topology contains the largest number of two-qubit interactions, it consistently exhibits the smallest effective support over the computational basis and maintains a lower absolute separation between class centroids than the other circuit families. At the same time, the centroid distance alone does not reproduce the progressive degradation observed at the kernel level, since it generally remains stable or even increases with the number of qubits. This indicates that the deterioration of the kernel cannot be explained solely by changes in the separation between class centroids, but instead reflects a broader reorganization of the embedding geometry. The relationship between the embedding metrics and the resulting kernel properties is examined jointly in the following subsection.

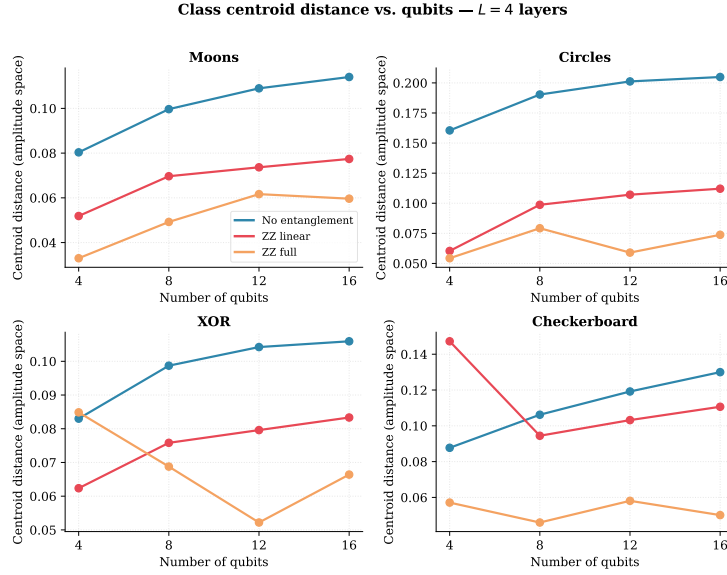


Figure 5.16: Distance between the class centroids in Hilbert space as a function of the number of qubits for the three circuit topologies with four layers. Source: own work.

5.5.3 Connecting kernel and embedding structure

The kernel and embedding analyses presented in the previous subsections provide two complementary views of the same phenomenon. While the kernel metrics describe how classification performance and generalization evolve as the circuits become larger, the embedding metrics reveal how the underlying quantum state changes with both the number of qubits and the circuit topology. Taken together, they provide a coherent picture of how the structure of the embedding influences the quality of the resulting kernel.

Figure 5.16 shows that the no-entanglement circuit preserves the separation between class centroids more effectively as the number of qubits increases. This behaviour is consistent with the kernel-level observations of Section 5.5.1, where the same architecture exhibits the smallest train–test gap, the lowest support vector fraction and the weakest signs of kernel concentration. Preserving the relative geometry of the classes in the embedding space therefore appears to produce kernels that require fewer support vectors and generalize more reliably.

This behaviour can be understood from the structure of the feature map itself. Without entanglement, each qubit evolves independently and depends only on a single input feature, so increasing the number of qubits mainly reproduces the same low-dimensional geometry in a larger Hilbert space. Introducing entanglement progressively mixes information across

qubits, generating increasingly complex correlations that modify this geometric structure. As the number of qubits grows, these correlations become more pronounced, and the embedding is consistently associated with the concentration effects observed at the kernel level. The results therefore suggest that the degradation is not caused by system size alone, but by the interaction between larger quantum systems and increasingly entangling circuit topologies.

This does not imply that entanglement should always be avoided. Figure 5.8 shows that the linear-entanglement circuit matches or outperforms the no-entanglement architecture on most datasets, with the difference being particularly evident for XOR, where the product-state embedding remains close to random guessing. A moderate amount of entanglement therefore appears beneficial for increasing the expressive power of the feature map and capturing more complex decision boundaries.

Overall, the results indicate that neither the number of qubits nor the entanglement topology should be considered in isolation when designing quantum feature maps. Increasing the size of the quantum system does not automatically improve the learned representation; instead, its effect depends strongly on how information is distributed throughout the circuit. Product-state embeddings preserve the original geometric relationships more effectively as the system grows, but may lack sufficient expressivity for some datasets. At the opposite extreme, dense entanglement increasingly amplifies concentration effects as the number of qubits grows, producing kernels with poor generalization despite high training performance. Within the circuit families considered in this Work, the linear-entanglement topology consistently occupies an intermediate regime, retaining part of the expressive benefit of entanglement while delaying the onset of severe kernel degradation. These observations suggest that the scalability of quantum feature maps depends not only on the number of qubits employed, but also on how those additional qubits are incorporated into the circuit architecture.

6. Conclusions

The objective of this Work was to analyse how different design choices in quantum feature maps affect both classification performance and scalability. Rather than focusing exclusively on predictive accuracy, the study examined how circuit architecture influences the geometry of the resulting embeddings and the structure of the induced quantum kernels.

The first experiment showed that the theoretical notions of expressibility and entangling capability proposed in the literature are only partially predictive of practical performance. Although expressibility exhibited a moderate positive correlation with classification accuracy, neither property consistently explained the behaviour observed across all datasets. These results indicate that global circuit descriptors alone are insufficient to characterize the effectiveness of quantum feature maps.

The second experiment demonstrated that the contribution of each circuit component depends strongly on the complexity of the learning task. Simple datasets could already be classified accurately using appropriate data encoding, whereas more challenging problems required the introduction of entangling operations. At the same time, increasing the amount of entanglement beyond a certain point did not systematically improve performance, showing that additional circuit complexity does not necessarily translate into better generalization.

The third experiment extended this analysis by studying how the different architectures scale with the number of qubits. Simpler feature maps continued to benefit from larger quantum systems, while the most expressive architectures progressively reached a regime where additional qubits produced diminishing or even negative returns in test accuracy despite continued improvements during training. These observations show that increasing the representational capacity of a quantum feature map is not, by itself, sufficient to obtain better predictive performance.

The final experiment provided an explanation for this behaviour by jointly analysing the kernel matrices and the corresponding quantum embeddings. Increasing both the number of qubits and the amount of entanglement produced progressively more concentrated kernels, characterized by lower off-diagonal variance, larger effective dimensions and higher support vector fractions. At the embedding level, the fully entangling circuits consistently occupied fewer effective basis states and maintained smaller separations between class centroids than the other architectures. Together, these results indicate that the observed

degradation originates from changes in the geometry of the embedding rather than from insufficient expressive power.

Overall, the results of this Work suggest that the design of effective quantum feature maps requires balancing expressivity with the preservation of meaningful geometric structure. While entanglement is essential for solving sufficiently complex classification problems, excessively dense entangling architectures progressively deteriorate the quality of the induced kernel as the system size grows. Within the families of circuits considered here, intermediate architectures consistently provided the best compromise between expressive power and generalization, highlighting that scalability depends not only on the number of qubits employed, but also on how quantum correlations are introduced throughout the circuit.

7. Future work

Although this Work provides a systematic analysis of several fundamental design choices in quantum feature maps, the space of possible circuit architectures remains considerably larger than the subset explored here. Designing an effective quantum feature map involves numerous interdependent decisions, including the choice of encoding gates, the arrangement of trainable rotations, the entanglement topology, the circuit depth and the number of qubits. Moreover, each of these components admits many alternative implementations, ranging from different gate sets and connectivity patterns to more sophisticated data-encoding strategies. Consequently, each design dimension constitutes a research direction in its own right, and no general prescription can yet be established for constructing optimal quantum feature maps.

An important limitation of the present study is that the trainable single-qubit rotations were initialized with random parameters that remained fixed throughout the experiments. This choice was made deliberately in order to isolate the effect of the architectural decisions themselves and to avoid introducing an additional optimization process that could mask their individual contributions. However, in practical applications these parameters can be optimized during training. Investigating how the different circuit families evolve once their variational parameters are learned constitutes one of the most relevant extensions of this Work. In particular, it would be interesting to determine whether parameter optimization mitigates the concentration effects observed in the more expressive circuits or, conversely, accentuates the differences between the architectural choices identified here.

Another natural extension is to broaden the experimental benchmark. Although the datasets considered in this Work cover several representative nonlinear classification problems, they remain relatively small and low-dimensional. Evaluating the proposed analyses on a larger and more diverse collection of datasets, including problems with different dimensionalities, sample sizes, noise levels and geometric structures, would make it possible to assess the robustness of the observed trends more thoroughly. Such a benchmark could also help identify whether particular circuit architectures are consistently better suited to specific classes of learning problems.

Ultimately, the long-term objective is not simply to compare individual circuit architectures, but to establish general design principles for quantum feature maps. Rather than relying on empirical trial and error, future research should aim to determine how the differ-

ent architectural choices interact and how they should be adapted to the characteristics of each learning problem and to the available quantum resources. Despite the rapid progress of quantum machine learning, the systematic design of quantum feature maps remains an open research problem. Developing such design methodologies therefore represents one of the most important challenges—and opportunities—for future work in the field.

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A. Additional experimental results

This Appendix contains the complete numerical results and supplementary figures that support the experiments presented in Chapter 5. These materials are included for completeness and reproducibility, while the main text focuses on the most relevant observations and trends.

A.1 Experiment 2

Table A.1 reports the complete test accuracy obtained for every circuit configuration evaluated in Experiment 2.

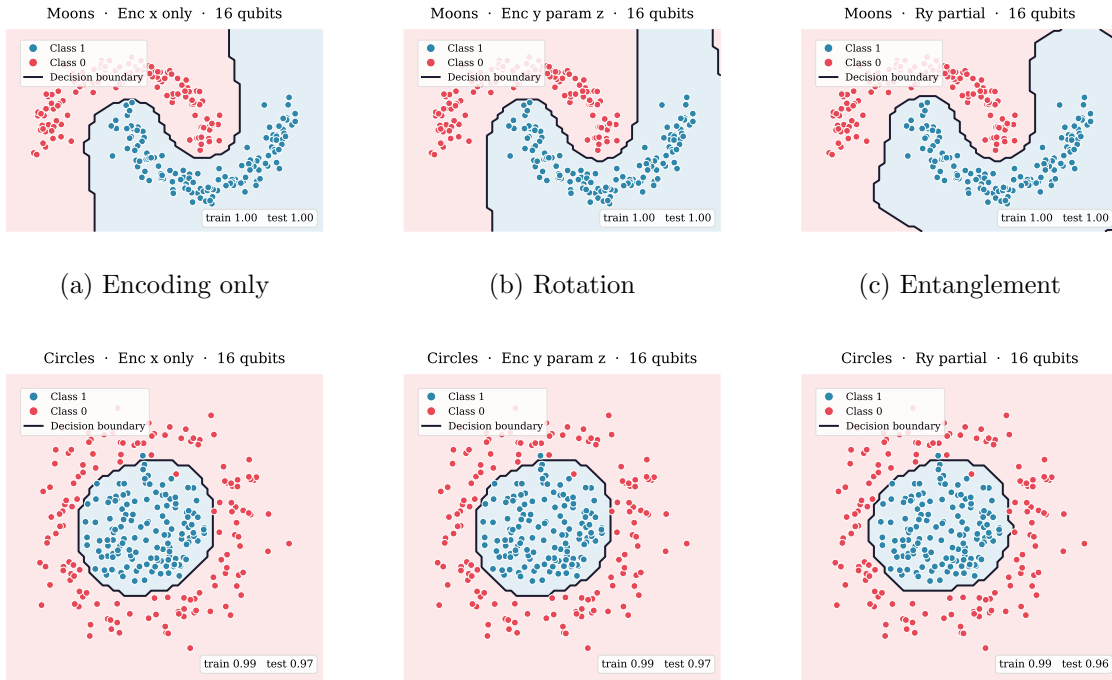
Table A.1: Test accuracy for encoding-only, rotation-axis combinations, and entanglement-level configurations across all datasets. Source: own work.

Configuration	Moons	Circles	Xor	Checkerboard
Encoding only (Block 1)				
enc_X_only	0.975	0.975	0.992	0.608
enc_Y_only	0.975	0.975	0.992	0.608
enc_Z_only	0.425	0.442	0.483	0.475
Rotation axis combinations (Block 1)				
enc_X_param_X	0.942	0.983	0.933	0.617
enc_X_param_Y	0.960	0.954	0.823	0.498
enc_X_param_Z	0.977	0.942	0.989	0.692
enc_Y_param_X	0.960	0.954	0.823	0.498
enc_Y_param_Y	0.942	0.983	0.933	0.617
enc_Y_param_Z	0.977	0.942	0.989	0.692
enc_Z_param_X	0.907	0.942	0.992	0.545
enc_Z_param_Y	0.907	0.942	0.992	0.545
enc_Z_param_Z	0.503	0.513	0.483	—
Entanglement levels (Block 2)				
full	0.998	0.983	0.991	0.848
partial	1.000	0.983	0.991	0.865

A.2 Experiment 3

Section 5.4 uses the Checkerboard dataset to illustrate how the decision boundaries evolve as the number of qubits increases, since it provides the clearest visual example of the effect of scaling the quantum system.

Figure A.1 presents the final decision boundaries obtained with the three circuit architectures on the Moons and Circles datasets using 16 qubits. Unlike the Checkerboard example, these datasets admit comparatively simple decision boundaries and all three architectures achieve very similar classification performance at the largest system size. Consequently, the resulting decision boundaries are also very similar, indicating that once sufficient expressive power has been reached, the choice of architecture has little visible effect on the learned classifier for these tasks.



Top row: Moons dataset.

Bottom row: Circles dataset.

Figure A.1: Decision boundaries obtained with the three circuit architectures using 16 qubits on the Moons (top row) and Circles (bottom row) datasets. Source: own work.

A.3 Experiment 4

This Appendix contains the complete numerical results together with the supplementary figures omitted from Section 5.5 for readability.

The discussion in the main text focuses on the representative case of $L = 4$, since it captures the behaviour observed throughout the experiment while avoiding unnecessary repetition. The additional figures included here show that the same qualitative trends are consistently reproduced for the remaining layer counts.

In particular, the kernel metrics exhibit the same progression towards concentration independently of the number of layers, while the embedding metrics preserve the same relative behaviour between circuit architectures. The complete decision-boundary visualizations likewise confirm that the fragmentation discussed in Section 5.5.1 is not specific to a particular layer configuration.

A.3.1 Complete classification results

Tables A.2–A.5 report the complete train and test accuracies for every circuit, dataset, qubit count and layer configuration evaluated in Experiment 4.

Table A.2: Training and test accuracies (mean $\pm$ standard deviation) for all circuit configurations, qubit counts, and datasets with $L = 1$. Source: own work.

Dataset	Circuit	Q=4		Q=8		Q=12		Q=16	
		Train	Test	Train	Test	Train	Test	Train	Test
Dataset: checkerboard									
	no_entanglement	0.939 ± 0.008	0.893 ± 0.045	0.946 ± 0.010	0.908 ± 0.021	0.951 ± 0.008	0.908 ± 0.021	0.959 ± 0.005	0.912 ± 0.020
	zz_full	0.714 ± 0.017	0.568 ± 0.056	0.855 ± 0.014	0.573 ± 0.024	0.929 ± 0.013	0.587 ± 0.021	0.973 ± 0.008	0.595 ± 0.023
	zz_linear	0.926 ± 0.009	0.830 ± 0.015	0.976 ± 0.003	0.828 ± 0.022	0.991 ± 0.005	0.845 ± 0.038	0.996 ± 0.003	0.847 ± 0.042
Dataset: circles									
	no_entanglement	0.711 ± 0.015	0.680 ± 0.035	0.733 ± 0.012	0.680 ± 0.036	0.756 ± 0.016	0.680 ± 0.044	0.770 ± 0.006	0.680 ± 0.038
	zz_full	0.717 ± 0.022	0.590 ± 0.031	0.844 ± 0.010	0.588 ± 0.033	0.921 ± 0.007	0.602 ± 0.050	0.966 ± 0.003	0.593 ± 0.039
	zz_linear	0.782 ± 0.020	0.638 ± 0.039	0.875 ± 0.023	0.648 ± 0.050	0.919 ± 0.011	0.673 ± 0.049	0.945 ± 0.007	0.673 ± 0.046
Dataset: moons									
	no_entanglement	0.814 ± 0.025	0.753 ± 0.038	0.826 ± 0.018	0.735 ± 0.016	0.831 ± 0.013	0.737 ± 0.022	0.839 ± 0.009	0.735 ± 0.014
	zz_full	0.733 ± 0.014	0.578 ± 0.029	0.876 ± 0.009	0.593 ± 0.025	0.936 ± 0.005	0.595 ± 0.041	0.971 ± 0.008	0.612 ± 0.038
	zz_linear	0.857 ± 0.019	0.747 ± 0.039	0.902 ± 0.007	0.755 ± 0.023	0.933 ± 0.003	0.752 ± 0.011	0.952 ± 0.009	0.747 ± 0.025
Dataset: xor									
	no_entanglement	0.586 ± 0.024	0.493 ± 0.052	0.605 ± 0.014	0.488 ± 0.062	0.629 ± 0.014	0.497 ± 0.060	0.653 ± 0.019	0.493 ± 0.062
	zz_full	0.777 ± 0.029	0.698 ± 0.043	0.896 ± 0.010	0.700 ± 0.017	0.946 ± 0.016	0.693 ± 0.033	0.966 ± 0.007	0.692 ± 0.025
	zz_linear	0.733 ± 0.013	0.610 ± 0.078	0.807 ± 0.010	0.620 ± 0.075	0.854 ± 0.009	0.622 ± 0.065	0.891 ± 0.008	0.630 ± 0.065

Table A.3: Training and test accuracies (mean $\pm$ standard deviation) for all circuit configurations, qubit counts, and datasets with $L = 2$. Source: own work.

Dataset	Circuit	Q=4		Q=8		Q=12		Q=16	
		Train	Test	Train	Test	Train	Test	Train	Test
Dataset: checkerboard									
	no_entanglement	0.679 ± 0.033	0.545 ± 0.017	0.839 ± 0.012	0.705 ± 0.054	0.856 ± 0.009	0.752 ± 0.040	0.869 ± 0.009	0.755 ± 0.035
	zz_full	0.712 ± 0.019	0.528 ± 0.032	0.919 ± 0.017	0.503 ± 0.037	0.990 ± 0.005	0.565 ± 0.038	0.999 ± 0.002	0.527 ± 0.034
	zz_linear	0.884 ± 0.007	0.713 ± 0.007	0.974 ± 0.011	0.782 ± 0.042	0.986 ± 0.006	0.787 ± 0.035	0.994 ± 0.003	0.780 ± 0.020
Dataset: circles									
	no_entanglement	0.716 ± 0.008	0.675 ± 0.026	0.736 ± 0.005	0.688 ± 0.021	0.756 ± 0.018	0.692 ± 0.023	0.766 ± 0.016	0.688 ± 0.014
	zz_full	0.699 ± 0.009	0.563 ± 0.017	0.913 ± 0.007	0.603 ± 0.029	0.996 ± 0.003	0.522 ± 0.032	1.000 ± 0.000	0.523 ± 0.042
	zz_linear	0.804 ± 0.021	0.665 ± 0.022	0.924 ± 0.006	0.680 ± 0.059	0.950 ± 0.007	0.677 ± 0.049	0.965 ± 0.007	0.672 ± 0.055
Dataset: moons									
	no_entanglement	0.736 ± 0.023	0.638 ± 0.029	0.791 ± 0.008	0.662 ± 0.024	0.811 ± 0.012	0.660 ± 0.019	0.824 ± 0.016	0.670 ± 0.026
	zz_full	0.721 ± 0.032	0.532 ± 0.058	0.924 ± 0.013	0.562 ± 0.023	0.990 ± 0.005	0.558 ± 0.029	1.000 ± 0.000	0.520 ± 0.027
	zz_linear	0.848 ± 0.006	0.685 ± 0.031	0.922 ± 0.013	0.707 ± 0.027	0.955 ± 0.005	0.708 ± 0.024	0.969 ± 0.006	0.708 ± 0.027
Dataset: xor									
	no_entanglement	0.579 ± 0.021	0.502 ± 0.050	0.611 ± 0.016	0.497 ± 0.066	0.647 ± 0.022	0.495 ± 0.061	0.674 ± 0.019	0.492 ± 0.050
	zz_full	0.733 ± 0.016	0.545 ± 0.007	0.936 ± 0.009	0.560 ± 0.025	0.989 ± 0.007	0.563 ± 0.036	1.000 ± 0.000	0.540 ± 0.046
	zz_linear	0.754 ± 0.009	0.613 ± 0.062	0.856 ± 0.006	0.622 ± 0.068	0.900 ± 0.017	0.630 ± 0.042	0.933 ± 0.011	0.630 ± 0.027

Table A.4: Training and test accuracies (mean $\pm$ standard deviation) for all circuit configurations, qubit counts, and datasets with $L = 4$. Source: own work.

Dataset	Circuit	Q=4		Q=8		Q=12		Q=16	
		Train	Test	Train	Test	Train	Test	Train	Test
Dataset: checkerboard									
	no_entanglement	0.639 ± 0.018	0.555 ± 0.017	0.725 ± 0.014	0.578 ± 0.015	0.793 ± 0.014	0.620 ± 0.037	0.832 ± 0.012	0.640 ± 0.028
	zz_full	0.724 ± 0.026	0.557 ± 0.074	0.931 ± 0.014	0.520 ± 0.037	0.996 ± 0.003	0.508 ± 0.026	1.000 ± 0.000	0.517 ± 0.028
	zz_linear	0.896 ± 0.015	0.767 ± 0.036	0.983 ± 0.006	0.775 ± 0.036	0.989 ± 0.004	0.748 ± 0.033	0.992 ± 0.003	0.710 ± 0.040
Dataset: circles									
	no_entanglement	0.678 ± 0.018	0.575 ± 0.010	0.724 ± 0.015	0.593 ± 0.027	0.762 ± 0.027	0.595 ± 0.029	0.790 ± 0.022	0.598 ± 0.025
	zz_full	0.705 ± 0.011	0.552 ± 0.043	0.937 ± 0.013	0.540 ± 0.053	0.999 ± 0.002	0.480 ± 0.054	1.000 ± 0.000	0.548 ± 0.029
	zz_linear	0.839 ± 0.025	0.617 ± 0.040	0.960 ± 0.010	0.663 ± 0.057	0.978 ± 0.009	0.658 ± 0.026	0.988 ± 0.004	0.665 ± 0.037
Dataset: moons									
	no_entanglement	0.750 ± 0.022	0.632 ± 0.032	0.795 ± 0.019	0.673 ± 0.055	0.819 ± 0.017	0.675 ± 0.040	0.837 ± 0.014	0.675 ± 0.031
	zz_full	0.723 ± 0.029	0.512 ± 0.050	0.947 ± 0.009	0.553 ± 0.058	0.999 ± 0.002	0.520 ± 0.017	1.000 ± 0.000	0.477 ± 0.039
	zz_linear	0.850 ± 0.019	0.660 ± 0.019	0.964 ± 0.011	0.703 ± 0.043	0.981 ± 0.003	0.688 ± 0.058	0.990 ± 0.006	0.682 ± 0.049
Dataset: xor									
	no_entanglement	0.590 ± 0.019	0.508 ± 0.046	0.622 ± 0.022	0.505 ± 0.054	0.644 ± 0.025	0.512 ± 0.055	0.663 ± 0.025	0.503 ± 0.057
	zz_full	0.735 ± 0.023	0.565 ± 0.053	0.939 ± 0.014	0.533 ± 0.035	0.999 ± 0.002	0.517 ± 0.035	1.000 ± 0.000	0.517 ± 0.048
	zz_linear	0.781 ± 0.011	0.598 ± 0.043	0.914 ± 0.014	0.612 ± 0.055	0.946 ± 0.011	0.610 ± 0.049	0.966 ± 0.004	0.630 ± 0.035

A.3.2 Additional kernel analysis

Figures A.2-A.5 present the kernel metrics for the remaining layer counts ($L = 1$, $L = 2$ and $L = 8$), complementing the representative $L = 4$ results discussed in the main text.

Across all layer configurations, the same qualitative behaviour is observed. Increasing the number of qubits reduces the off-diagonal variance of the kernel matrix while increasing

Table A.5: Training and test accuracies (mean $\pm$ standard deviation) for all circuit configurations, qubit counts, and datasets with $L = 8$. Source: own work.

Dataset	Circuit	Q=4		Q=8		Q=12		Q=16	
		Train	Test	Train	Test	Train	Test	Train	Test
Dataset: checkerboard									
	no_entanglement	0.644 ± 0.020	0.537 ± 0.045	0.730 ± 0.014	0.565 ± 0.041	0.770 ± 0.009	0.580 ± 0.035	0.806 ± 0.018	0.587 ± 0.032
	zz_full	0.716 ± 0.015	0.460 ± 0.046	0.946 ± 0.008	0.520 ± 0.015	0.996 ± 0.004	0.503 ± 0.020	1.000 ± 0.000	0.538 ± 0.052
	zz_linear	0.869 ± 0.023	0.618 ± 0.052	0.984 ± 0.006	0.738 ± 0.045	0.992 ± 0.003	0.685 ± 0.060	0.997 ± 0.003	0.632 ± 0.046
Dataset: circles									
	no_entanglement	0.679 ± 0.018	0.583 ± 0.034	0.739 ± 0.024	0.568 ± 0.034	0.776 ± 0.010	0.572 ± 0.015	0.814 ± 0.009	0.577 ± 0.018
	zz_full	0.711 ± 0.027	0.542 ± 0.030	0.946 ± 0.006	0.530 ± 0.027	0.999 ± 0.002	0.477 ± 0.061	1.000 ± 0.000	0.493 ± 0.030
	zz_linear	0.839 ± 0.021	0.573 ± 0.050	0.989 ± 0.005	0.587 ± 0.048	0.996 ± 0.003	0.625 ± 0.037	0.997 ± 0.002	0.603 ± 0.066
Dataset: moons									
	no_entanglement	0.684 ± 0.019	0.590 ± 0.032	0.778 ± 0.023	0.650 ± 0.036	0.830 ± 0.017	0.672 ± 0.035	0.848 ± 0.018	0.680 ± 0.035
	zz_full	0.733 ± 0.011	0.502 ± 0.029	0.942 ± 0.008	0.492 ± 0.040	0.998 ± 0.003	0.517 ± 0.052	1.000 ± 0.000	0.495 ± 0.044
	zz_linear	0.847 ± 0.007	0.610 ± 0.038	0.985 ± 0.005	0.655 ± 0.043	0.995 ± 0.005	0.657 ± 0.041	0.997 ± 0.003	0.632 ± 0.034
Dataset: xor									
	no_entanglement	0.620 ± 0.029	0.475 ± 0.056	0.671 ± 0.022	0.473 ± 0.050	0.709 ± 0.014	0.473 ± 0.059	0.736 ± 0.015	0.482 ± 0.053
	zz_full	0.740 ± 0.020	0.557 ± 0.022	0.944 ± 0.011	0.498 ± 0.033	0.998 ± 0.003	0.503 ± 0.057	0.999 ± 0.002	0.507 ± 0.052
	zz_linear	0.824 ± 0.022	0.582 ± 0.014	0.967 ± 0.003	0.635 ± 0.051	0.984 ± 0.004	0.622 ± 0.049	0.990 ± 0.007	0.633 ± 0.036

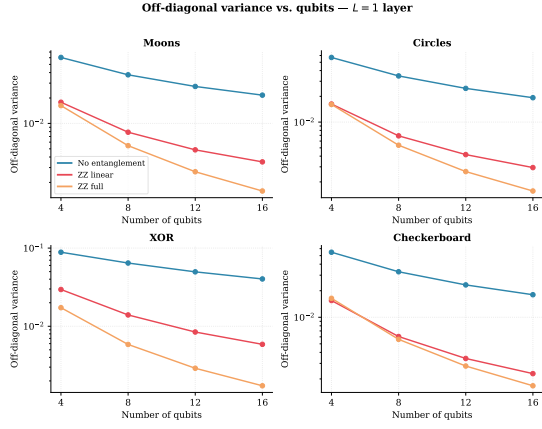
its effective dimension and the fraction of support vectors, accompanied by a widening train–test gap for the entangling architectures. These additional figures therefore reinforce that the kernel concentration effects described in Section 5.5.1 are systematic rather than specific to a particular number of layers.

A.3.3 Additional embedding analysis

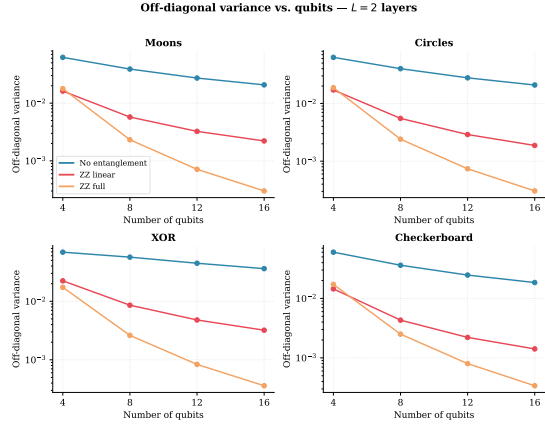
The embedding analysis presented in the main text is likewise illustrated using $L = 4$. Figures A.6–A.8 show the corresponding results for the remaining layer counts.

The additional entropy, participation-ratio and centroid-distance curves reproduce the same relative ordering between circuit architectures observed for $L = 4$. In particular, the no-entanglement circuit remains largely insensitive to the number of qubits after normalization, whereas the two entangling topologies exhibit genuine collective behaviour. The $L = 1$ entropy plot additionally illustrates the exact degeneracy discussed in Section 5.5.2, where all three circuits attain the maximum normalized Shannon entropy because the circuit consists solely of an initial Hadamard layer followed by diagonal phase operations.

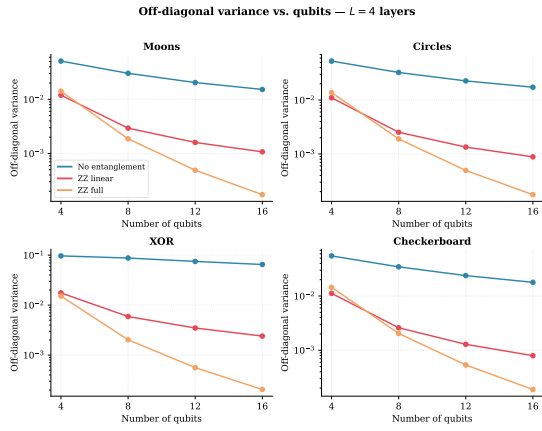
Finally, Figure A.9 presents the complete basis-state density profiles for every dataset and layer count at 16 qubits. These visualizations provide a complementary view of the embedding distributions discussed in the main text, highlighting the broader distributions produced by the linear-entanglement topology and the closer similarity between the no-



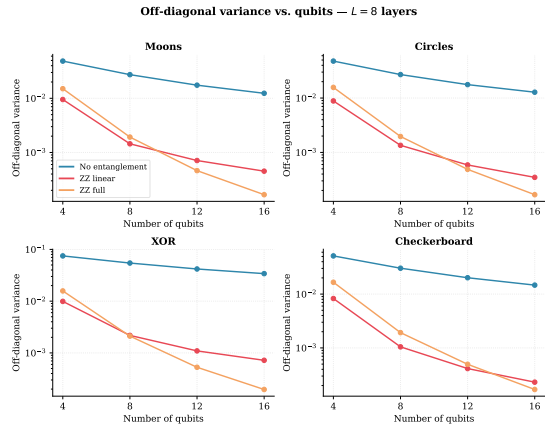
(a) Off-diagonal variance for $L = 1$.



(b) Off-diagonal variance for $L = 2$.



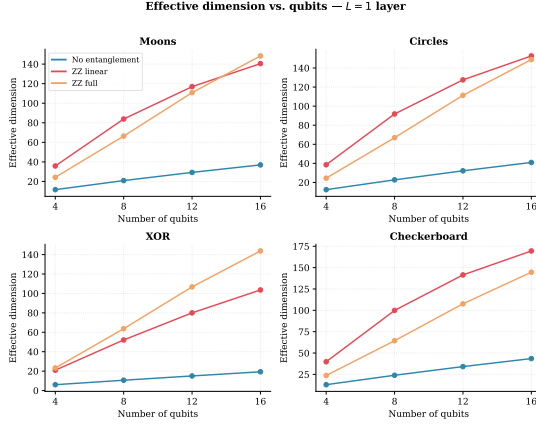
(c) Off-diagonal variance for $L = 4$.



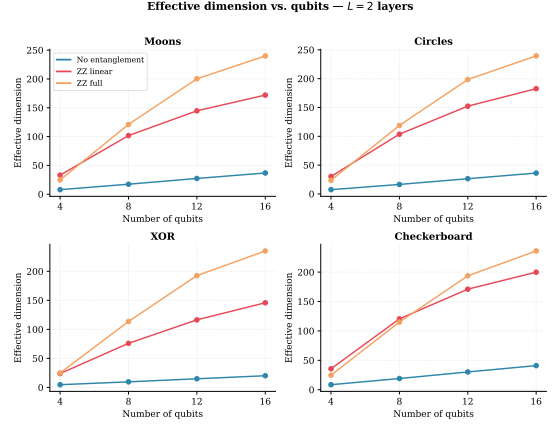
(d) Off-diagonal variance for $L = 8$.

Figure A.2: Off-diagonal variance of the kernel matrix for all circuit architectures as a function of the number of qubits and circuit depth. Source: own work.

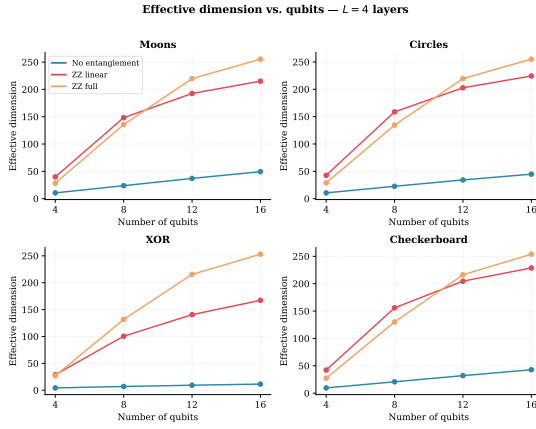
entanglement and full-entanglement circuits.



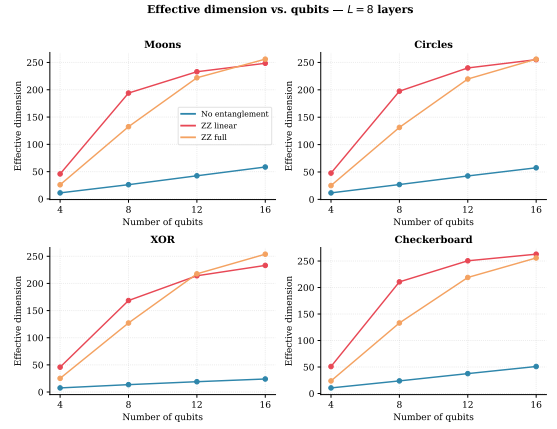
(a) Effective dimension for $L = 1$.



(b) Effective dimension for $L = 2$.

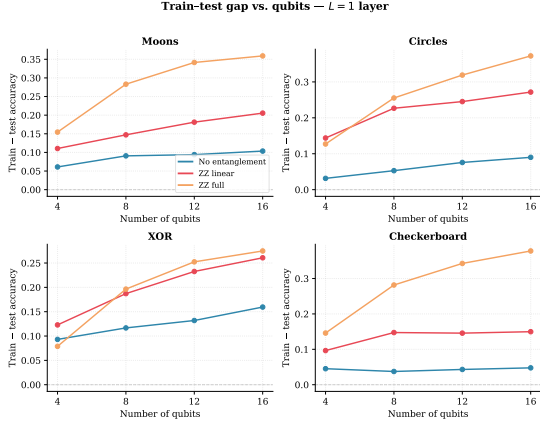


(c) Effective dimension for $L = 4$.

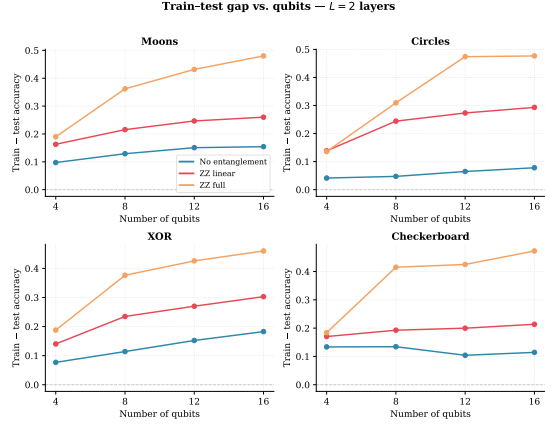


(d) Effective dimension for $L = 8$.

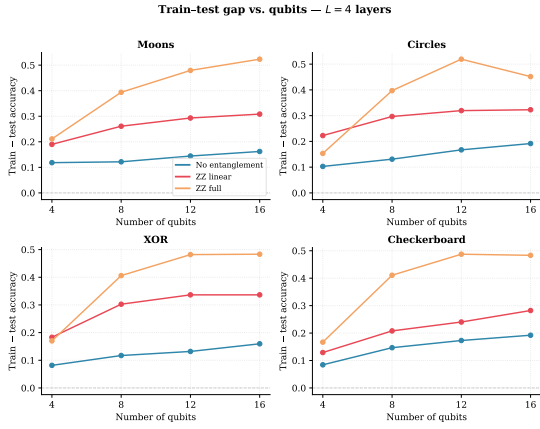
Figure A.3: Effective dimension of the kernel matrix for all circuit architectures as a function of the number of qubits and circuit depth. Source: own work.



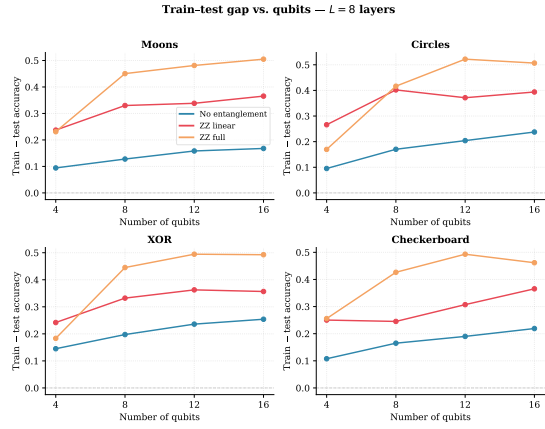
(a) Train-test gap for $L = 1$.



(b) Train-test gap for $L = 2$.

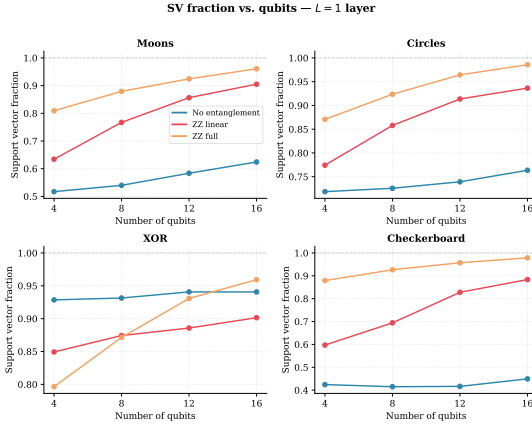


(c) Train-test gap for $L = 4$.

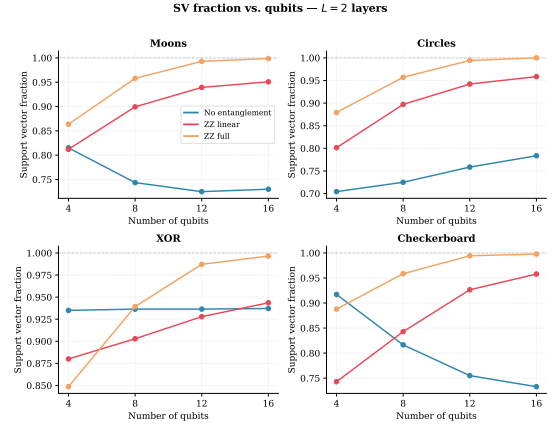


(d) Train-test gap for $L = 8$.

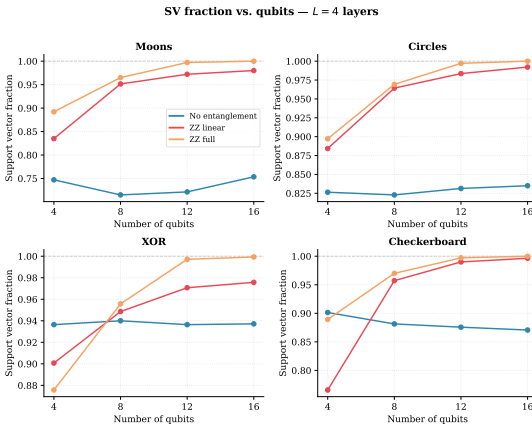
Figure A.4: Difference between training and test accuracy for all circuit architectures as a function of the number of qubits and circuit depth. Source: own work.



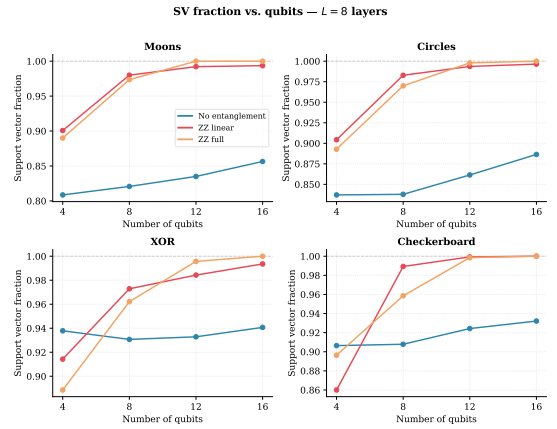
(a) Support vector fraction for $L = 1$.



(b) Support vector fraction for $L = 2$.

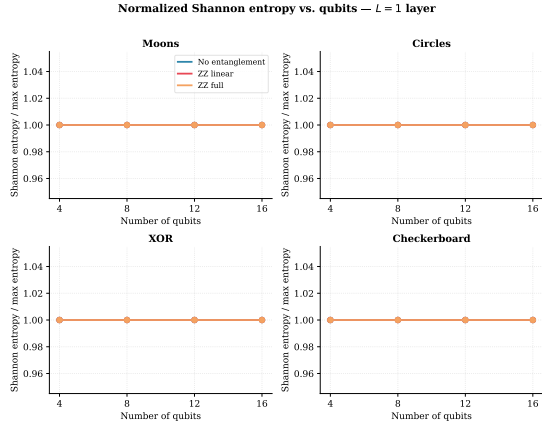


(c) Support vector fraction for $L = 4$.

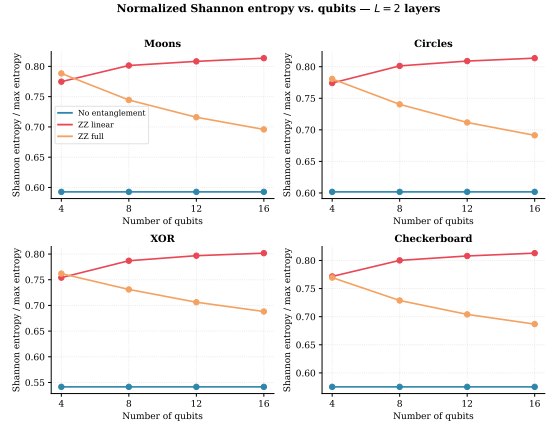


(d) Support vector fraction for $L = 8$.

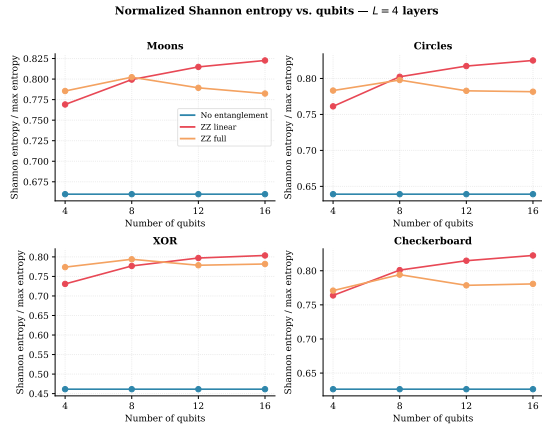
Figure A.5: Fraction of training samples selected as support vectors for all circuit architectures as a function of the number of qubits and circuit depth. Source: own work.



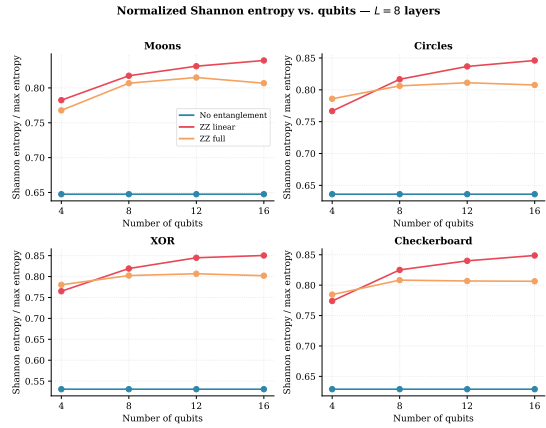
(a) Normalized Shannon entropy of circuits with $L=1$ and growing amount of qubits.



(b) Normalized Shannon entropy of circuits with $L=2$ and growing amount of qubits.



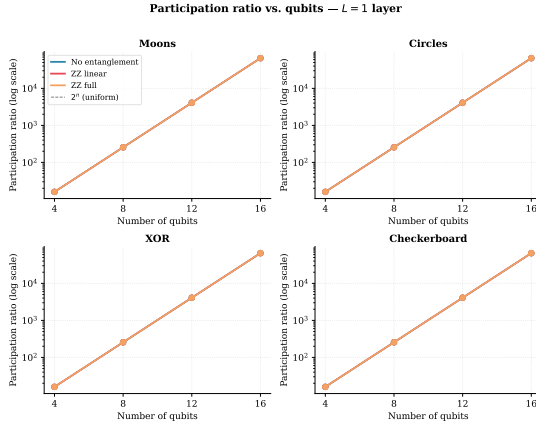
(c) Normalized Shannon entropy of circuits with $L=4$ and growing amount of qubits.



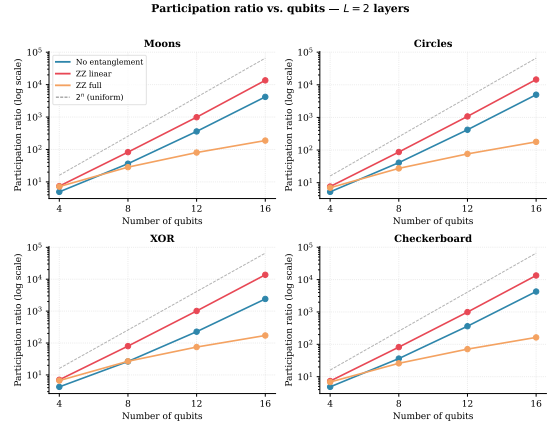
(d) Normalized Shannon entropy of circuits with $L=8$ and growing amount of qubits.

Figure A.6: Normalized Shannon entropy of all circuits with growing amount of qubits.

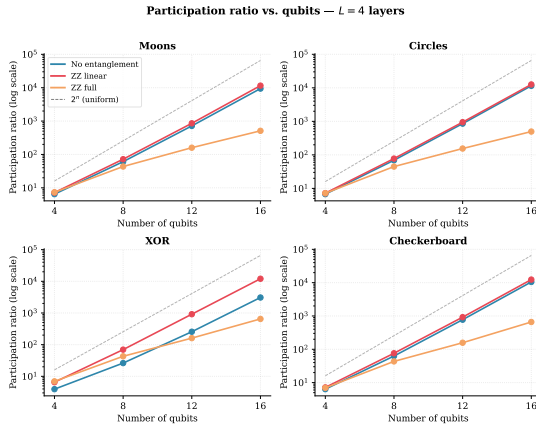
Source: own work.



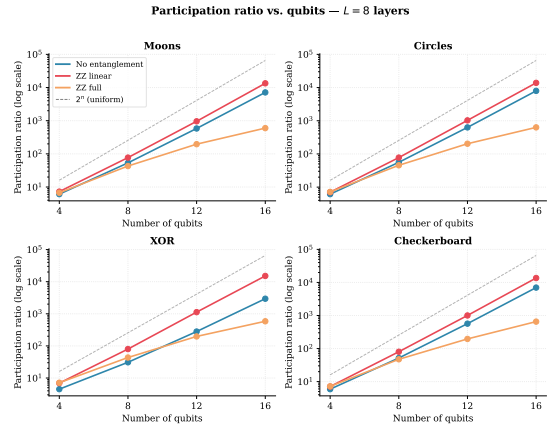
(a) Participation ratio of circuits with $L=1$ and growing amount of qubits.



(b) Participation ratio of circuits with $L=2$ and growing amount of qubits.

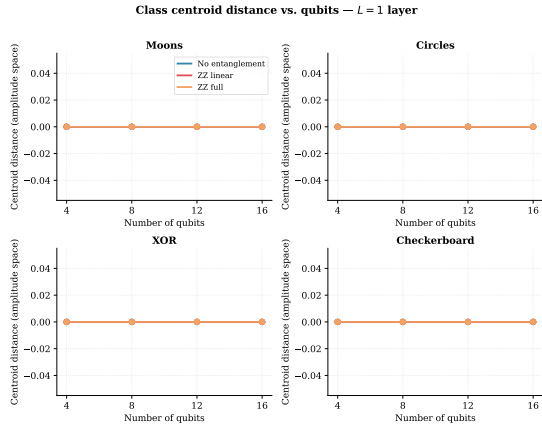


(c) Participation ratio of circuits with $L=4$ and growing amount of qubits.

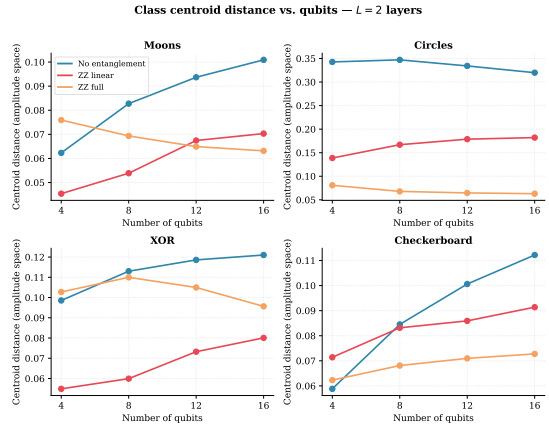


(d) Participation ratio of circuits with $L=8$ and growing amount of qubits.

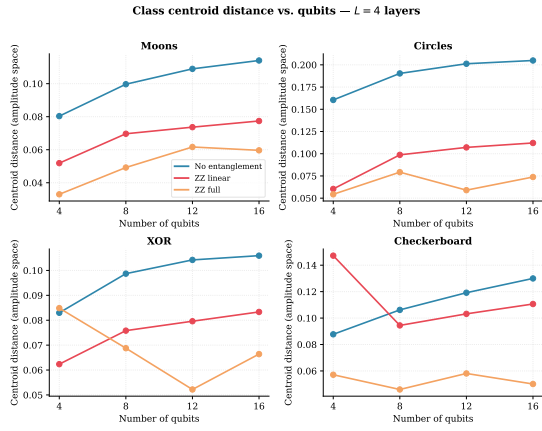
Figure A.7: Participation ratio of all circuits with growing amount of qubits. Source: own work.



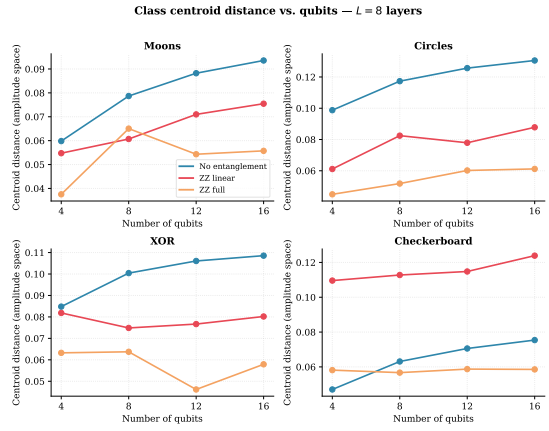
(a) Class centroid distance of circuits with $L=1$ and growing amount of qubits.



(b) Class centroid distance of circuits with $L=2$ and growing amount of qubits.



(c) Class centroid distance of circuits with $L=4$ and growing amount of qubits.



(d) Class centroid distance of circuits with $L=8$ and growing amount of qubits.

Figure A.8: Class centroid distance of all circuits with growing amount of qubits. Source: own work.

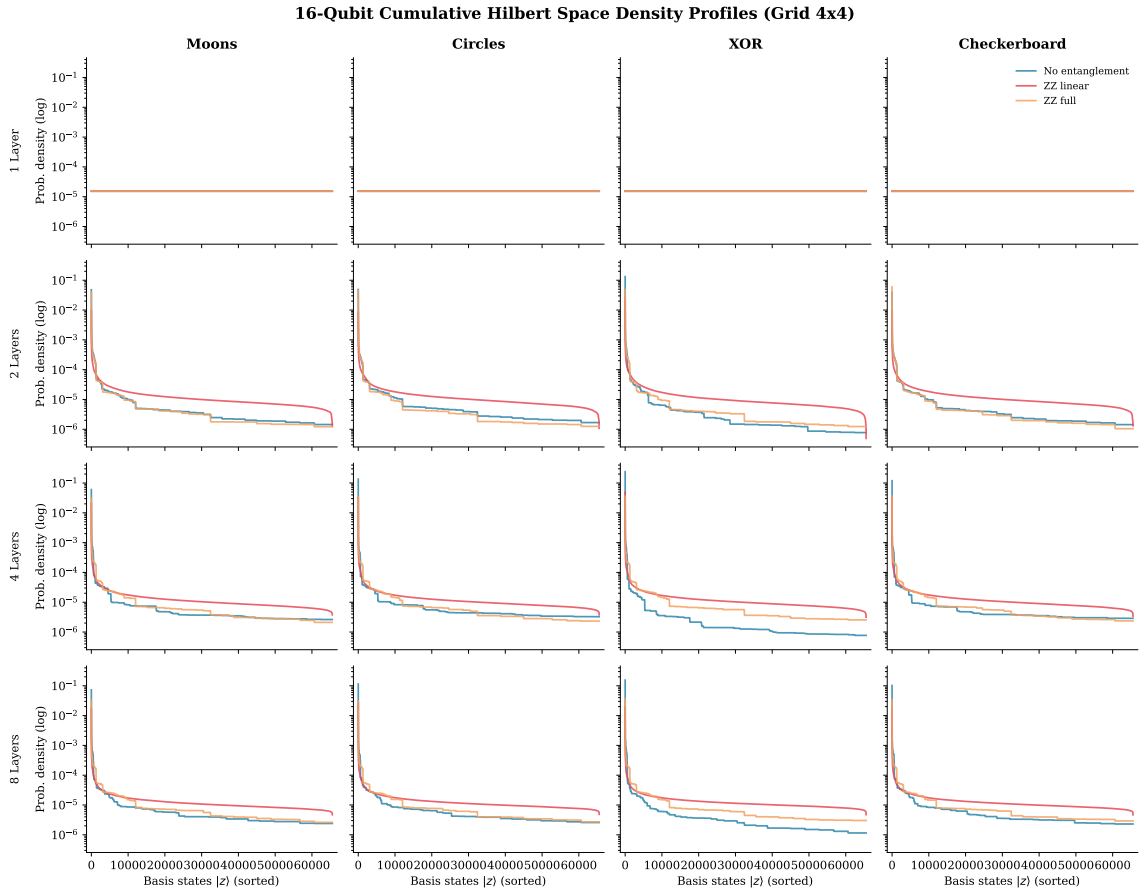


Figure A.9: Basis state density profiles at 16 qubits, for all four layer counts (rows) and all four datasets (columns). Source: own work.